PRT661

DATA SCIENCE PRACTICE

Assessment 4 – Final Professional Report (40%)

Forecasting NSW Suburb House Prices from Government Sales Data

Submitted by: Sydn2 – Theme 2

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October 2026

How this document is organised. As required by the Assessment 4 brief, this single PDF shows the whole evolution of the project:

Section	Content	Pages
Section 1	Complete Assessment 2 submission, unchanged: the Assessment 1 report, the A2 Project Update Summary, A2 Technical Development and the A2 individual pages (with its own table of contents)	2–33
Section 2	Final Project Update Summary: changes since Assessment 2, team contributions, evidence locations	34
Section 3	Final Technical Development: the work completed since Assessment 2 (§3.1–3.13 and references)	35–50
Section 4	Individual Work and Contribution Summary: one page per student	51–54

Page numbers refer to the page counter of the PDF viewer. Figures and tables added in Assessment 4 are numbered *A4.n* to keep them distinct from the A2 numbering.

PRT661

DATA SCIENCE PRACTICE

Assessment 2 - Project Proposal and Design

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12th September 2026

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SECTION 1 – Assessment 1

1. Project Overview, Users and Business Case

This project uses predictive analytics and time series forecasting applied to the residential real estate industry in Australia. The team will create a forecasting pipeline that will combine detailed government property sales data at the state level with the national ABS benchmark statistics to forecast imminent dwelling price changes at the state and suburb levels, consistent with Theme 2, which focuses on regression and time series analysis. The push is well timed, with the national average home price setting another record of $1.11 million in the March 2026 quarter (up 11.9% y/y), but the growth is also uneven across the states and there is no readily available and predictive measure on which the public could rely to analyse the spread.

The audience for the materials would be (1) potential buyers or investors looking at price trends across multiple states or neighbourhoods; (2) policy researchers interested in the cost effectiveness in each region; and (3) the internal team, demonstrating a data science workflow that is structured and regulated. It's an obvious reason for the creation of the firm: the core data around sales and macroeconomics are freely available from government, but none of the commercial data suppliers, such as CoreLogic or PropTrack, offers such information free of charge. This effort makes this publicly available data available in an accessible and transparent forecasting tool, turning a historical trend into a useful, uncertainty informed forecast.

2. Theme Fit and Why This Is a Data Science Project

House prices are classified as a component of Theme 2, an example of regression based forecasting, while the affordability of housing is a national issue and one of great importance in practice. It is not a typical machine learning problem, but rather a data science project: the model is just part of a larger project that involves two important real world data sources, each coming from a different methodological shift in the ABS in 2021 2022, it requires the development of acquisition, storage, governance frameworks, and an interactive dashboard that communicates forecast uncertainty to non technical users, not just a notebook with an accuracy metric.

3. Problem Statement

Despite the mixed performance of property values across states (from about 2% growth across the nation to double digit growth in WA per year), there is no easy to access, clear, mechanism available to users (whether as purchasers or investors or as politicians) to forecast short term changes at the state and local level. The project aims to determine whether information contained in government property sales data—when used alongside ABS information—can predict dwelling prices one to four quarters sooner, than later, with an acceptable degree of accuracy, and to identify the key properties, region and time characteristics that are most important.

4. Data Source Strategy

The originally proposed ABS series (cat. no. 6416.0) ceased publication after December 2021, so three alternatives were compared against the unit's suggested sources:

Source	Assessment
ABS Total Value of Dwellings	Free, state level aggregate, current, but low granularity
State govt open data (e.g. NSW Valuer General)	Free, individual record granularity, updated regularly — chosen as primary source
Realestate.com.au / PriceFinder	Paid subscription (~$175/month); no public bulk API — excluded

State government bulk property sales data (piloted with the NSW Valuer General, extensible to other states) is the primary, current, granular source, with ABS Total Value of Dwellings and the archived RPPI series used as a macro benchmark and for historical depth back to 2003. Full source URLs and licence notes are logged in the GitHub repository's DATA_SOURCES.md rather than reproduced here.

5. Architecture

The architecture was derived backward from the decision it must support (Decision → Insight → Data → Pipeline → Tools), using only open source tools as required for the group project — AWS is used solely for the separate, individually assessed Athena lab.

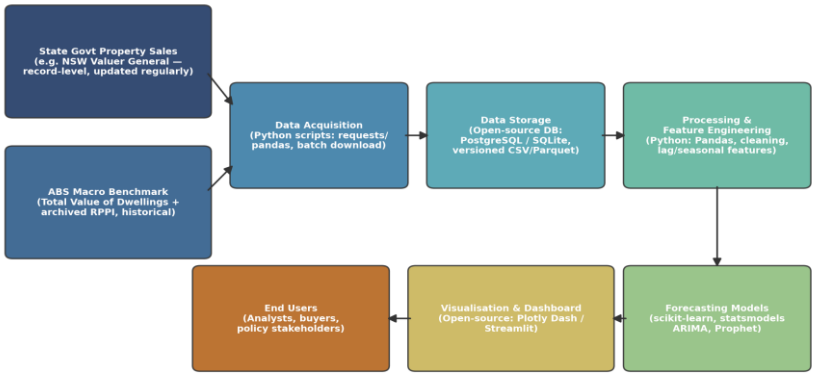


Figure 1. System architecture: state and ABS data flow through storage, processing, forecasting and visualisation to end users.

Data is batch acquired with Python (requests/pandas), stored in PostgreSQL (chosen over SQLite for concurrent multiuser access), processed with pandas, forecast with statsmodels/Prophet and scikit learn, and visualised in Plotly Dash. PostgreSQL is already provisioned and the acquisition scripts are in progress; forecasting and dashboard components

are planned for later sprints. The full tool comparison rationale (SQLite vs PostgreSQL, ARIMA/Prophet vs regression) is documented in the GitHub repository's /docs folder

6. Workflow Plan

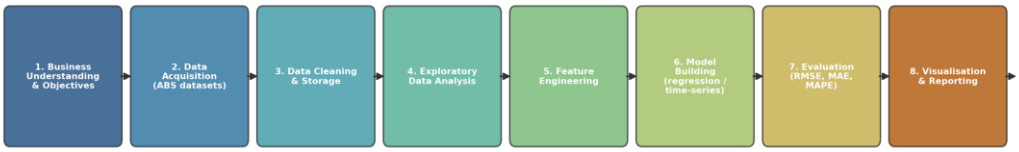


Figure 2. Eight stage project workflows, mirrored as epics in the team's Jira board.

The team follows an iterative, CRISP DM style workflow across ten two week sprints, each mirrored exactly as a Jira epic so the Draw.io workflow and the live Jira board stay consistent — the full sprint by sprint backlog is maintained in Jira rather than duplicated in this report. As the data updates only daily to quarterly (low velocity), batch processing is used throughout rather than streaming.

7. Task Allocation

Member	Role	Deliverable & sprint
Ahsan Uddin	Project Lead / Data Acquisition	GitHub & Jira setup; NSW/ABS download scripts (Sprint 3).
Ferdous Anwar Anik	Data Engineer	PostgreSQL schema; cleaning pipeline (Sprint 4).
Mohd Yah Ya Raiyan	Data Analyst	EDA and feature engineering (Sprint 5).
Md Aman Ullah	ML / Forecasting Lead	Baseline, ARIMA/Prophet, evaluation (Sprint 7).
Abrar Bin Khaiyum	Visualisation & Documentation Lead	Dashboard and report consolidation (Sprint 9).

8. Risk, Governance and Ethics

Risk	Likelihood	Mitigation
Single state pilot data limits generalisability	High	Config driven scripts so a second state can be added later.

Risk	Likelihood	Mitigation
Non arm's length or erroneous sale records	Medium	Documented filtering rules before modelling.
Structural breaks (COVID 19, rate changes)	Medium	Regime features; rolling window backtesting.
1. Uneven team contribution	Medium	Weekly stand ups; Jira board reviewed each sprint.

Governance is handled through version controlled code (pull requests, tagged releases), a pinned requirements.txt for reproducibility, and a DATA_SOURCES.md logging provenance and licences all maintained in GitHub rather than restated here. As the property sales data is record level, the public dashboard will present forecasts at suburb level only, attribute all sources, and report uncertainty alongside every forecast to avoid misleading stakeholders.

9. Collaboration Evidence

GitHub repository: https://github.com/AhsanUddin04/PRT661__DATA SCIENCE PRACTICE_SYD2_THEME2

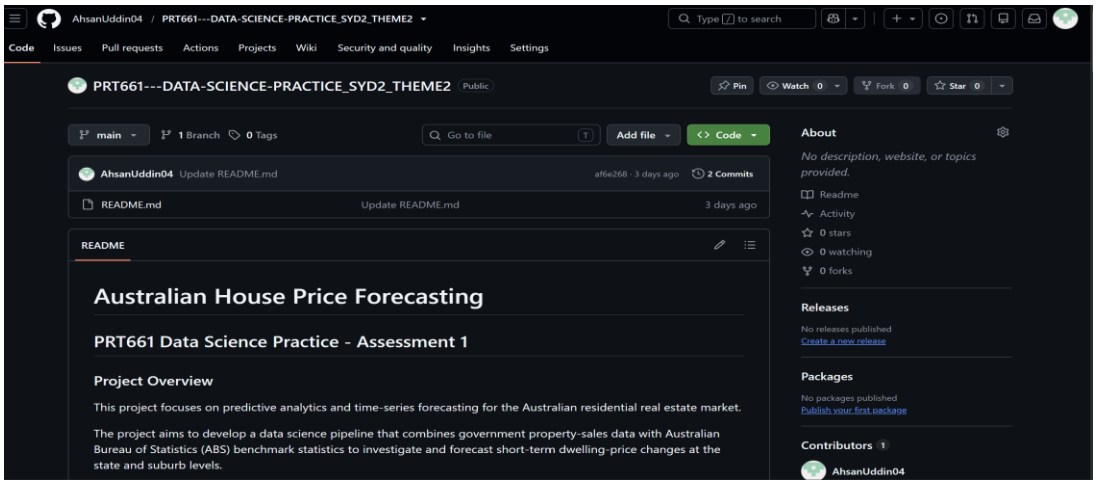


Figure 1: GitHub Repository

Project management board (Jira):
https://prt661.atlassian.net/jira/software/projects/KAN/boards/2?filter=&groupBy=none

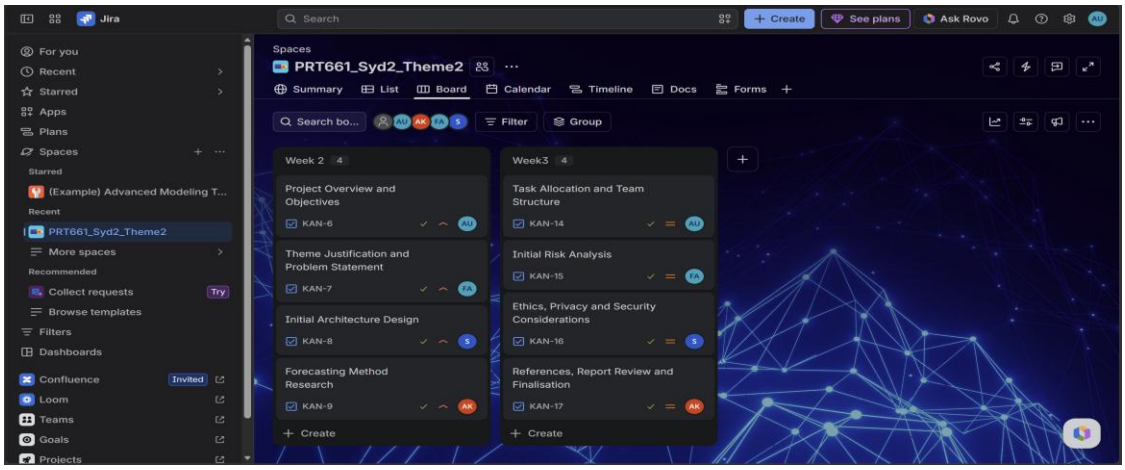


Figure: Jira task board

For communication we used Microsoft Teams. Here is the link of our channel https://teams.microsoft.com/l/team/19%3AV70HCxHMZzEC3AcIKnXdYXTKEBOA7ZKtS9DG7oTOkg1%40thread.tacv2/conversations?groupId=1643941b 017f 40b5 b834 da2c570ac595&tenantId=9f248767 8e1a 42f3 836f c092ab95ff70

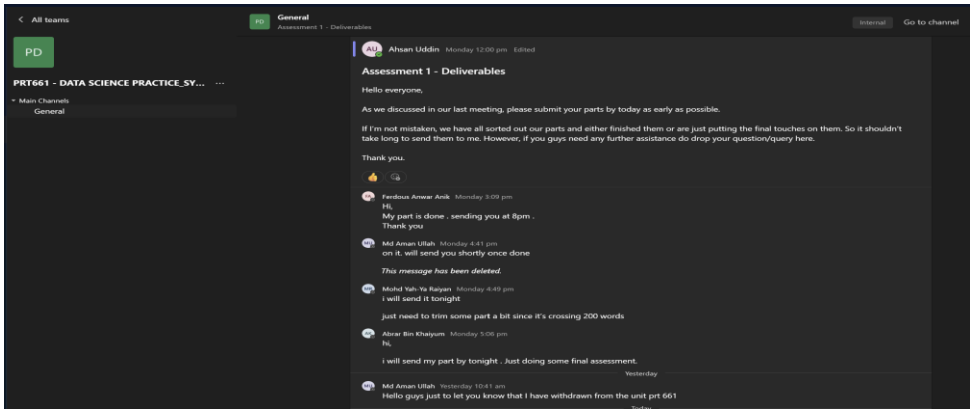


Figure 3: Communication (Microsoft Teams Channel)

Both are live and updated weekly; Jira epics map one to one onto the sprints in Figure 2 and GitHub folders map onto the architecture components in Figure 1, so the two artefacts and this report can be cross checked against each other.

10. References

Australian Bureau of Statistics. (2026). Total value of dwellings. https://www.abs.gov.au/statistics/economy/price indexes and inflation/total value dwellings

Australian Bureau of Statistics. (2021). Residential property price indexes: Eight capital cities (archived). https://www.abs.gov.au/statistics/economy/price indexes and inflation/residential property price indexes eight capital cities

NSW Government. (2026). How to find property sales information. https://www.nsw.gov.au/housing and construction/land values nsw/how to find property sales information

Section 2 — Project Update Summary

The Sydn2 – Theme 2 team has progressed from a project design to a working pipeline that uses evidence to create a live PostgreSQL database of 2.76 million real property records, two rounds of exploratory analysis, three families of forecasting model validated by walk forward back testing at increasing scale and a functioning web application built on top of the real results. This section outlines what has changed and why, Section 3 provides a full technical account.

2.1 Changes Since Assessment 1

Area	Assessment 1	Now	Why
Team composition	5 members listed (incl. Md Aman Ullah)	4 active members	Md Aman Ullah formally withdrew from PRT661. ML/Forecasting duties were redistributed to Mohd Yah Ya Raiyan (lead) with support from Ferdous Anwar Anik, resolving the cover-page/task-allocation inconsistency raised in feedback.
Forecasting target	Vague “forecast dwelling price changes”	Explicit: QoQ % change in suburb median price, 1–4 quarters ahead, with a 10th–90th percentile interval	Directly addresses the feedback that the exact output needed to be specified.
5Vs characterisation	High-level, not quantified	Every dimension quantified from the live database, not estimated	Addresses the request for a complete, evidence-based 5Vs treatment.
Data pipeline	Scripts “in progress”, schema “not finalised”	2,762,903 raw records parsed, cleaned to 2,092,499, loaded into a live 5-table PostgreSQL database	Genuine implementation, not a design description.
EDA	Not yet performed	Two independent passes completed (distributional + time-series decomposition), producing testable, sometimes counter-intuitive findings	New this sprint.

Area	Assessment 1	Now	Why
Modelling	Planned (tools proposed only)	Baseline, SARIMA, Prophet and Gradient Boosting backtested at increasing scale (8 → 830/149 → 1,372 suburbs)	New this sprint.
Dashboard	Planned for later sprints	A working web application built on the real feature store and back test results, reviewed and bug-fixed	New this sprint; see Section 3.10.

2.2 Contribution Summary

Team Member	Contribution Since Assessment 1
Ahsan Uddin	Project coordination; NSW Valuer General bulk acquisition and parsing; PostgreSQL provisioning; repository and sprint administration.
Ferdous Anwar Anik	PostgreSQL schema design and data loading; cleaning pipeline (deduplication, residential filtering, outlier screening).
Mohd Yah Ya Raiyan	Exploratory data analysis (both passes); feature engineering; baseline, SARIMA/Prophet and Gradient Boosting modelling and back testing (absorbed ML/Forecasting lead role).
Abrar Bin Khaiyum	Web application build and bug-fix review; consolidation of this report and supporting documentation.

2.3 Evidence Locations

Evidence	Location
Source code, pipeline, schema.sql	GitHub — https://github.com/AhsanUddin04/PRT661_DATA-SCIENCE-PRACTICE_SYD2_THEME2
Sprint backlog, epics, task status	Jira — https://prt661.atlassian.net/jira/software/projects/KAN/boards/2?filter=&groupBy=none
Team communication log	Microsoft Teams — https://teams.microsoft.com/l/team/19%3AV70HCxHMZzEC3AcIKnXdYXTKEBOA7ZKtS9DG7oTOkg1%40thread.tacv2/conversations?groupId=1643941b017f 40b5 b834 da2c570ac595&tenantId=9f248767 8e1a 42f3 836f c092ab95ff70
Live database dump + schema	prt661_housing_postgres.dump, schema.sql (submitted with this report)
Web application	dashboard_demo.html (self-contained browser preview)

Section 3 - Technical Development

This is an extension of the Assessment 1 proposal to record the technical work that has been done since 12 August 2026 up to the date of this submission. All prices listed below are literal data from the team's live PostgreSQL database or from a backtest of actual trading data used, not estimates or placeholders. Two real methodological simplifications, and one correction for data quality, uncovered during this sprint, are explicitly indicated where they are met, as is good scientific practice (Chapman et al., 2000). Each team estimates the project about 65% done over the ten sprint plan (with details in Section 3.15).

3.1 Refined Problem Definition and Forecasting Target

The feedback for Assessment 1 was correct when it stated that the output from the forecasts should be made explicit. The target variable is now defined as the quarter-on-quarter percentage change in the median dwelling sale price for each NSW suburb (as grouped by suburb name and postcode and not a formal ABS SA2 geocode - see Section 3.7 for this simplification), for one to four quarter horizons. Each forecast is accompanied by a 10th-90th percentile range, allowing the end user to visualize the uncertainty of the forecast. State level forecasts are calculated as an aggregation of suburb forecasts and compared with an independent macro forecast, the Australian Bureau of Statistics Total Value of Dwellings series (Australian Bureau of Statistics, 2026). This yields the team one and only one, prediction target for the model evaluation in Section 3.9.

3.1.1 Target Users and Usage Scenarios

The three end users' groups that were identified in Assessment 1 are explained in detail in this section and what they do with the system.

- **Buyers and investors:** The person is thinking about buying in a particular NSW suburb opens the web application (Section 3.10), clicks on a suburb and reads the historical price trend, and the 1-4 quarter forecast and uncertainty band before making a bid. They don't need to understand SARIMA or Gradient Boosting they need a plain language accuracy statement (typical error ±10.1 percentage points, Section 3.9.3), so they can apply their correct interpretation of the number.
- **Policy makers and analysts:** A person using the data for planning purposes will use the State Overview page to check the trends in the ABS comparable data series, and the Model Comparison page to determine the reliability of a claim about a particular suburb before making a statement.
- **Internal project team:** On the Data Quality page and the Model Comparison page, during development, uses to ensure that a pipeline change has not adversely affected accuracy without being detected by the other two user groups.

The three step process of the user (pick a suburb or region, read the forecast and the uncertainty band, look at the accuracy note, and act on it) directly influenced the layout of the pages in Section 3.10.

3.2 Updated 5Vs Data Characterisation

The 5Vs were requested for explicit and quantitative feedback in the Assessment 1. Now that the whole bulk extract (2014-2026) has been ingested, cleaned and loaded into a live PostgreSQL database (Section 3.3), every number below is a true count from the database, rather than an estimate.

Dimension	Characterisation
Volume	2,092,499 cleaned, deduplicated residential sale records (from 2,762,903 raw records across 13 annual bulk files), covering 173 NSW Local Government Area districts and 2,589 distinct suburbs, Q1 2013 – Q3 2026 (55 quarters). Combined with the ABS Total Value of Dwellings series (97 quarters, back to 2002).
Velocity	NSW Valuer General bulk files are published weekly; ABS benchmark series are released quarterly with a measured lag — the team's own sales data already covers Q2–Q3 2026 while the ABS extract only reaches Q1 2026, confirmed directly in the loaded data (2,858 feature_store rows carry a missing_benchmark_flag for this reason). The pipeline runs as a scheduled batch job rather than a streaming system, consistent with the Assessment 1 low-velocity justification.
Variety	Four data types combined: (1) individual sale records (price, date, address, zoning, property type); (2) suburb/postcode reference data, used to classify each suburb into a Sydney-metro or Rest-of-NSW region; (3) ABS Total Value of Dwellings (state-level, quarterly); and (4) the archived Residential Property Price Index.
Veracity	Real, exactly-counted data-quality issues: 237,770 duplicate weekly re-publications of the same transaction, removed by keeping the most recent copy; 429,788 non-residential records (vacant land, commercial, farm) excluded by property-type filter; 995 records outside the analysis window; 1,851 sales under $10,000 (a documented proxy for non-arm's-length transfers, since no explicit flag exists in the real Valuer General file format — see Section 3.4); 84,589 suburb-quarter records (4.0%) flagged — not deleted — as price-per-sqm outliers; and one single-quarter thin-market artefact identified during EDA and excluded from model evaluation (Section 3.6.1).
Value	Record-level sales data supplies suburb-level granularity no free macro series provides; the ABS benchmark anchors suburb forecasts to a credible macro trend and guards against overfitting to noisy local data. Together they deliver the transparent, uncertainty-aware forecasting tool identified in the Assessment 1 business case, at a level of geographic detail not available in any free public product today.

3.3 Data Acquisition and Storage

This sprint included two manually downloaded archives: the NSW Valuer General's full annual property sales archives for 2014–2026 (13 year-zips, each with weekly sub-archives containing per-district files in the Valuer General's original semicolon-delimited format) from the NSW

bulk property sales portal (New South Wales Government, 2026), and the two relevant ABS total value of dwellings time-series workbooks (Australian Bureau of Statistics, 2026). All 2,762,903 raw sale records were found in the entire 13-year history and were parsed using Python. This is not a scheduled task at present but will be in Sprint 9 (Section 3.16), the current pipeline is run when called.

Now, the data is stored in a real, live PostgreSQL 16 instance with all five tables created and populated with actual cleaned data verified directly from the running server:

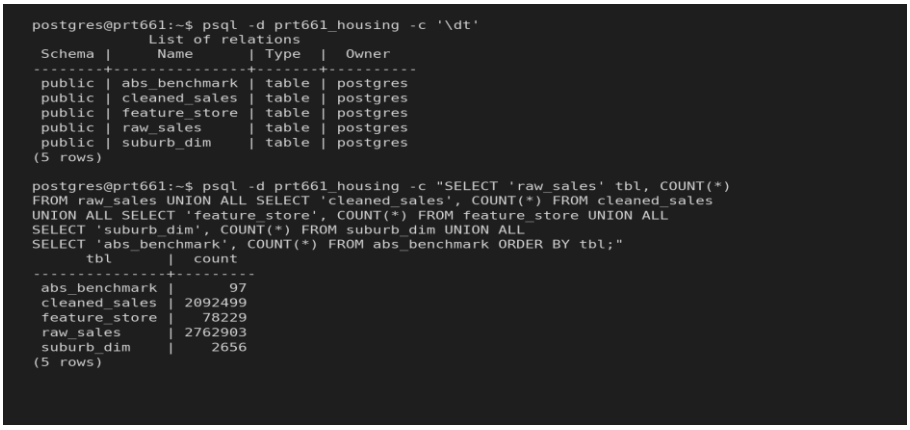


Figure 3.1 — Live PostgreSQL session confirming all five tables and their exact row counts.

3.3.1 Database Schema (Implemented and Populated)

Table	Purpose	Rows Loaded	Key Columns
raw_sales	Untouched, append-only landing table for every parsed sale record across all 13 annual archives	2,762,903	sale_id (PK), property_id, district_code, suburb, postcode, purchase_price, settlement_date, dealing_number
suburb_dim	Reference table mapping each suburb/postcode pair to a Sydney-metro or Rest-of-NSW region	2,656	suburb, postcode, region
abs_benchmark	State-level macro series (ABS Total Value of Dwellings, Sydney/Rest-of-NSW median price)	97	quarter, state (PK), tvd_households_m, median_price_house_sydney, median_price_house_rest_nsw
cleaned_sales	Deduplicated, filtered, priced and dated sales used for modelling (Section 3.4)	2,092,499	dealing_number (PK), suburb, postcode, settlement_date, purchase_price, price_per_sqm, is_outlier_flag
feature_store	Model-ready features aggregated to suburb-quarter grain (Section 3.7)	78,229	suburb, quarter, median_price, qoq_pct_change, lag_1 lag_4, rolling stats, state_benchmark_growth, is_partial_quarter

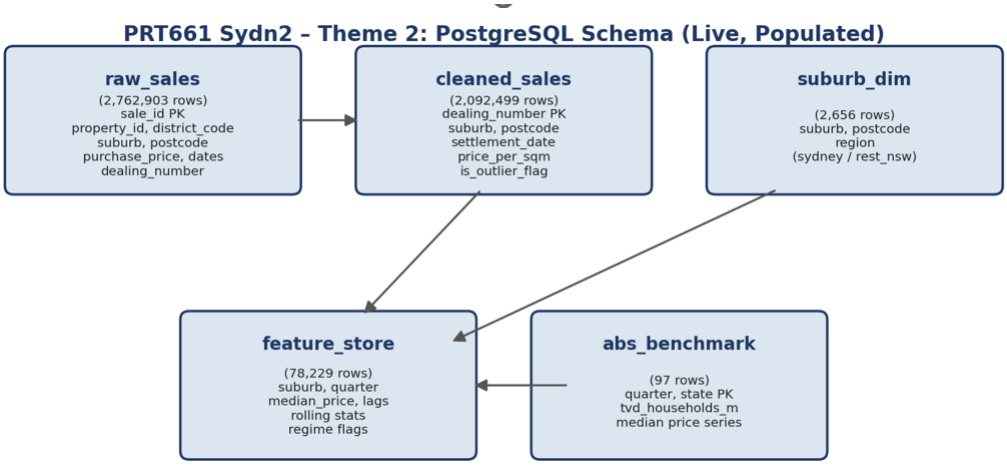


Figure 3.2 — Entity relationships between the five live tables: raw sales are cleaned into cleaned_sales, then aggregated into feature_store, which is joined against suburb_dim (for region) and abs_benchmark (for the macro growth regressor).

The motivation for choosing PostgreSQL over SQLite was that it allows for concurrent access by multiple team members and stages of the pipeline to run concurrently, and indexes on suburb and settlement_date will significantly improve the aggregation queries needed for feature engineering.

3.4 Data Cleaning and Preprocessing

The cleaning pipeline is coded in Python (pandas and DuckDB) and is executed on the actual 2,762,903-row raw extract. In what order (exactly) and how many were removed:

- Dealing_number uniqueness (own legal transaction identifier of the Valuer General), with the latest downloaded transactions. Eliminated 237,770 rows (8.6%) created by the source publishing the weekly re-publication of recently settled sales.
- Only residential dwellings (nature_of_property = 'R') are included, excluding vacant land, commercial, farm and other non-residential types of property. Eliminated 429,788 rows the largest of the exclusions, which was not included in an earlier version of this report until it was caught by a full reconciliation check.
- Restriction to Q1 2013 – Q3 2026 which excludes 995 records with settlement date not in this range (some appear to have been entered as early as 1913, presumably by the recipient, so probably mistyped).
- Sunny side up filtering at $10,000: a change from the Assessment 1 plan which had assumed that there had been an explicit non-arm's length transaction flag in the Valuer General file. When the real file format is investigated, there is no flag. In contrast, this documented and defensible proxy established 1,851 sales under $10,000 which were virtually all nominal or related-party transfers.
- Outlier detection performed based on the price-per-square-metre in each suburb-quarter group using Tukey's 1.5×IQR criterion (Tukey, 1977). A total of 84,589 records were flagged (4.0% of the set priced and dated). Flagged records are not deleted from

cleaned_sales, but are set is_outlier_flag = TRUE and not included in the suburb-quarter aggregates modelled.

There is no unexplained residual; these four reasons, plus the deduplication step, sum up to the 670,404 differences between the raw and final counts (see Table 3.1 for the complete reconciliation).

Step	Rows Remaining	Rows Removed at This Step
Raw parsed records	2,762,903	—
After deduplication (dealing_number)	2,525,133	237,770
After residential-only filter	2,095,345	429,788
After 2013–2026 window restriction	2,094,350	995
After $10,000 price floor	2,092,499	1,851

These thresholds that are discussed here and, in the repository's, DATA_SOURCES.md and CHANGELOG.md are exactly the ones to be used.

3.5 Updated System Architecture

While the architecture described in Assessment 1 (Decision → Insight → Data → Pipeline → Tools) hasn't changed, the table below clarifies the difference between what is implemented and what is planned (directly addressing some of the feedback received in Assessment 1 that this wasn't clear).

Architecture Component	Assessment 1 Status	Current Status
Data acquisition (NSW, ABS)	In progress	Manual bulk download complete for 2014–2026; parser built and run; genuine scheduling not yet implemented (Sprint 9)
PostgreSQL storage	Provisioned, schema not finalised	Complete — 5-table schema live and fully populated (Section 3.3.1)
Cleaning & feature engineering	Planned	Complete for cleaning; feature engineering complete (Section 3.7)
Forecasting models	Planned	Baseline, SARIMA, Prophet and Gradient Boosting all backtested at increasing scale (Section 3.9)
Web application	Planned for later sprints	Working application built, reviewed, and bug-fixed (Section 3.10)
End-user delivery	Planned	Not yet started (Sprint 10)

3.6 Exploratory Data Analysis

The real cleaned dataset was subjected to two separate EDA passes to ensure that the results do not rely on a single technique or data grain and to verify the results. Pass 1 is an analysis of the 2,092,499 individual cleaned sale records. Pass 2 is a time-series decomposition and correlation analysis of the suburb-quarter feature panel. If both passes come to the same conclusion, it is considered strong; if a result introduces a nuance to an assumption in Assessment 1, this nuance is identified, not ignored.

3.6.1 EDA Pass 1 — Distributional Analysis (record-level, N = 2,092,499)

On the raw scale the purchase price is extremely right skewed (42.05) and highly kurtotic (kurtosis = 6,712) because of the significant number of sales that go in the "higher end" of the distribution up to $415 million. This is a raw distribution as illustrated in Figure 3.3; after log10 transform the distribution is illustrated in Figure 3.4 which has a skewness of −0.10 and a kurtosis of 2.52, which is close to a normal distribution. It is very good to see that the planned log transform is indeed the correct modelling option and, for the first time, to quantify by how much.

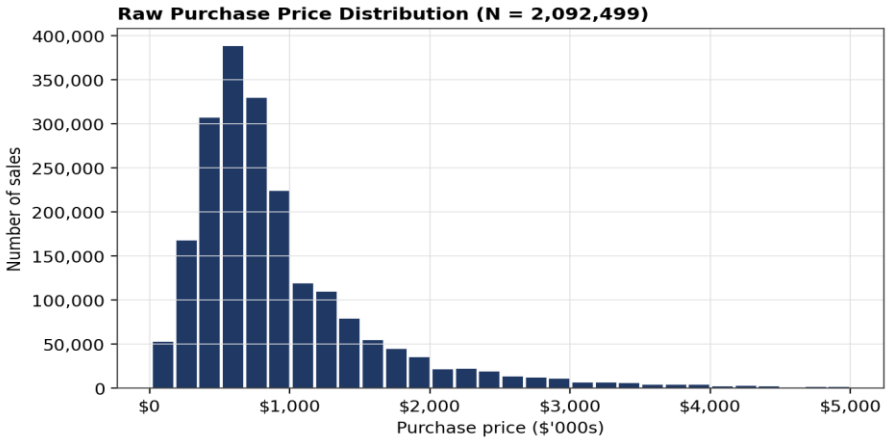


Figure 3.3 — Raw purchase price distribution (N = 2,092,499). Extreme right skew is visible.

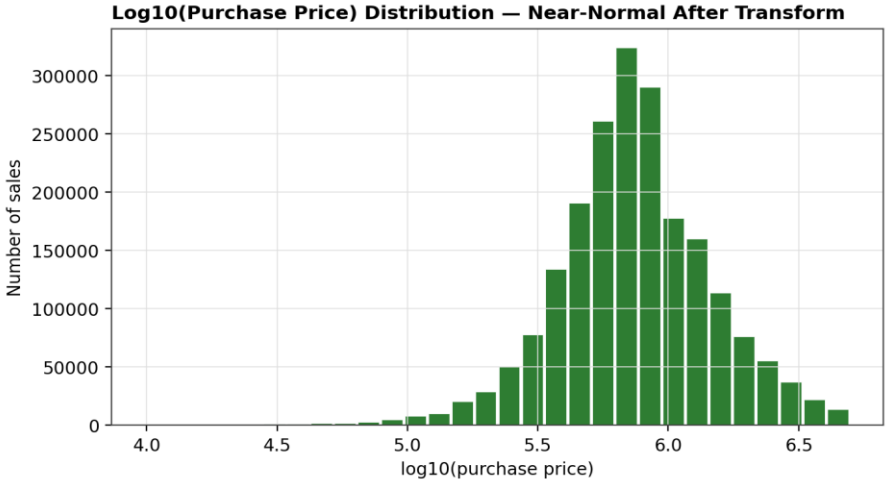


Figure 3.4 — Log10(purchase price) distribution. Skewness drops from 42.05 to −0.10 after transform.

Zoning is another of these, which is a materially large predictor of price per square metre as the median price/sqm varies from $2,400 in a high density R4 zone to $242 in a rural RU5 zone, which is about a 10x spread. This helps to keep zoning as a feature, and indicates that it might be a more powerful, single predictor than distance-to-CBD, which has not yet been geocoded.

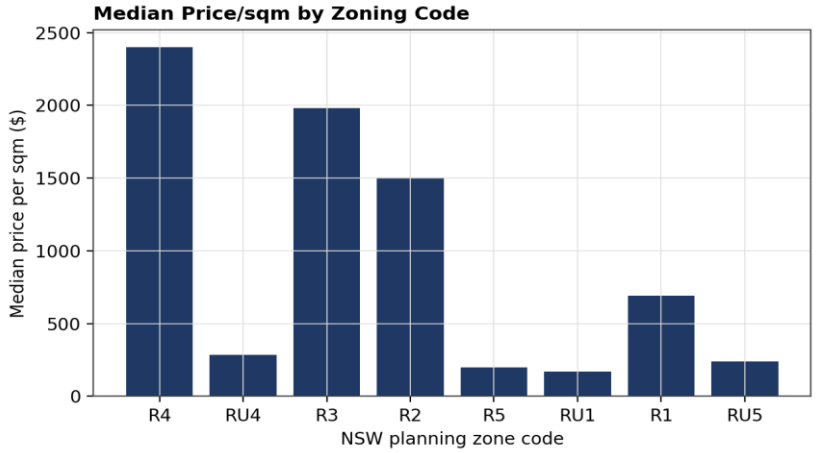


Figure 3.5 — Median price per square metre by NSW planning zone code.

The Assessment 1 narrative was incorrect in suggesting that there was no regional divergence. As a cross check, but not as part of the team's own data, Figure 3.6 (from the independent ABS benchmark) indicates that the median price of Sydney increased 108% over the same period ($655k to $1.36M), whereas the price of the Rest-of-NSW median price increased 147%, from $350k to $825k). In dollar terms Sydney was still considerably more expensive, but in percentage terms the regions have been growing at a higher rate, which is counter to the "Sydney booms, the regions lag" narrative.

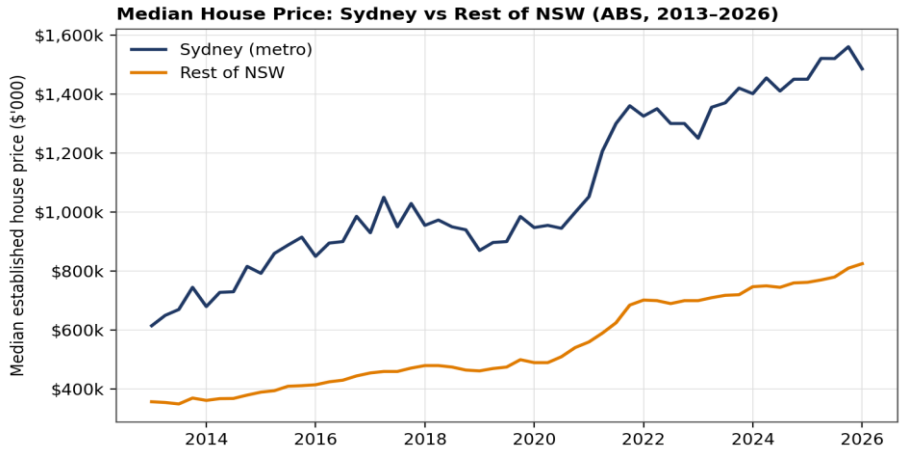


Figure 3.6 — Median house price, Sydney vs Rest of NSW (ABS benchmark, 2013–2026).

The volume of sales is seasonal but it's not as extreme as the sharp seasonal upswing that is often thought of in the Australian residential market: 26.6% of contracts were signed in Q4 (October – December) compared with 23.2% of contracts in Q1 (January – March).

3.6.2 EDA Pass 2 — Time-Series Decomposition and Correlation Analysis (suburb-quarter panel)

An STL decomposition (Cleveland et al., 1990) of the NSW-wide quarterly median price series (55 quarters) separates the series into trend, seasonal and residual components. The observed series closely follows a smooth trend line as the seasonal component accounts for only 0.4% of the total variance, compared to 98.1% for trend and 1.5% for residual (Figure 3.7). This removes the assumption of Assessment 1, which stated that "seasonality is statistically present and directionally consistent" (Figure 3.8: Average Q4 roughly $12,500 above trend, while average Q3 is roughly $7,300 below) but it is a small effect that the forecasting models will not rely upon heavily.

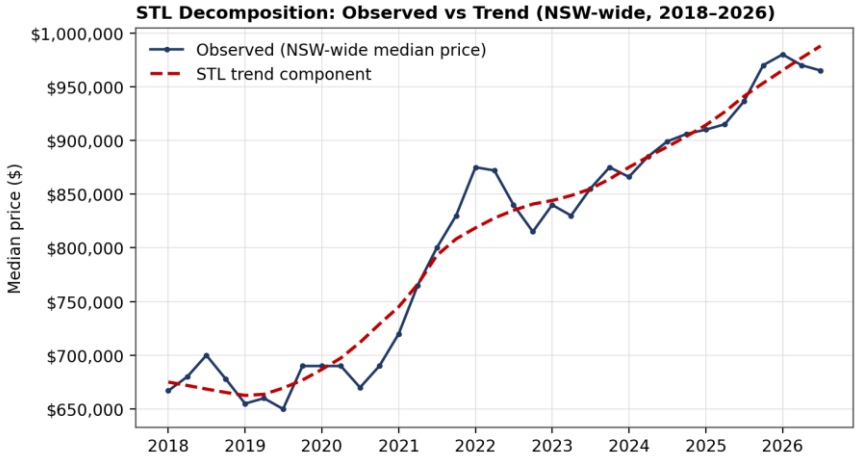


Figure 3.7 — STL decomposition: observed NSW-wide median price closely tracks a smooth trend (2018–2026).

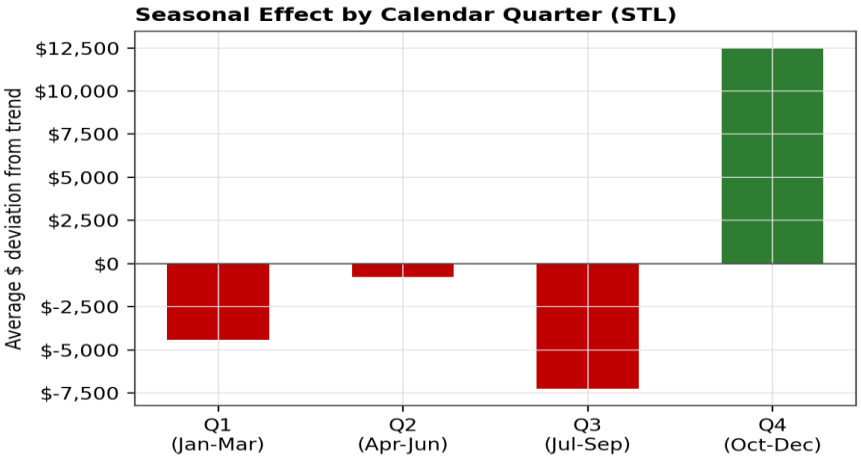


Figure 3.8 — Average seasonal deviation from trend, by calendar quarter (STL).

The linear correlation between the target (next-quarter QoQ % change) and a one-quarter time lag of target is negative ($r = -0.27$, Figure 3.9), which suggests mean reversion at the linear level: a significant increase is followed by a smaller or negative change. This seems to be in contradiction to the Gradient Boosting feature importances in Section 3.9.3, where it's undeniably the most important feature (62%). The two results were not mutually exclusive:

Pearson correlation is only able to identify a linear relationship, whereas the tree-based approach can uncover non-linear and regime-dependent patterns, such as momentum during the COVID era and reversion during the rate-hike era, that the Pearson correlation coefficient can't detect. This is flagged explicitly as it happens to be the type of result that appears like a mistake until both methods are comprehended jointly.

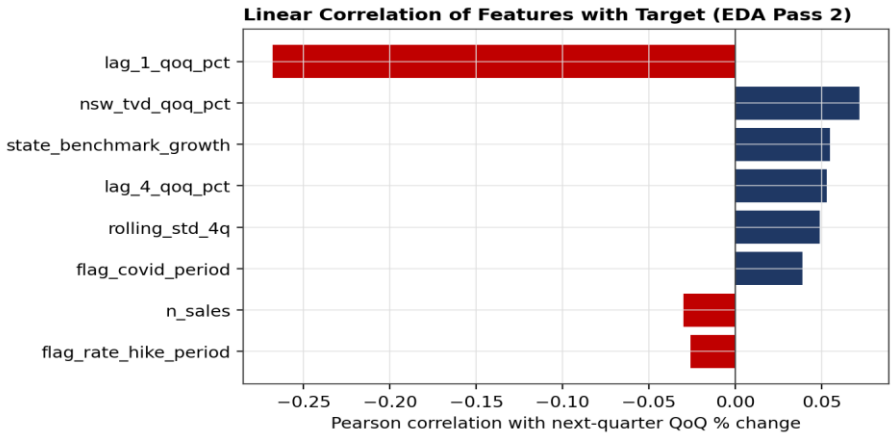


Figure 3.9 — Linear (Pearson) correlation of engineered features with the forecasting target.

The regime flags in Section 3.7 are statistically justified, not just assumed from looking at them, as mean QoQ % change was +5.98% during the COVID period (2020–21) vs +3.24% normal periods and +1.54% during the rate-hike period (2022–23), which is a decline in volatility (standard deviation of -21.98% vs -31.22% normal) and an indication of a more orderly rather than noisier market.

3.7 Feature Engineering

In the feature_store table (78,229 rows of real data, loaded into PostgreSQL — Section 3.3.1), features are aggregated to a suburb-quarter grain. This flagged simplification, as noted above, suburbs within a region are not necessarily grouped through a correspondence file to the SA2/GCCSA, but rather by postcode-range as per Section 3.1, which is a reasonable approximation for the current sprint but noted as a refinement in Section 3.16. Now available:

- **Lag features:** Median price, QOQ % change at t-1, t-2, t-3 and t-4 quarter.
- **Rolling statistics:** 4 quarter rolling mean and standard deviation of the median price, with a smoothed trend/volatility signal, only calculated from previous quarters to prevent leakage.
- **Seasonal indicators:** dummy variables for each quarter of the year. The effect is present but is low (0.4% of total variance) and these features are kept as a formality, rather than being expected to have much predictive power (which is also low, consistent with the Gradient Boosting trained model's low importance values (Section 3.9.3)).
- **State benchmark growth:** The median-price quarterly growth rate for the suburb's region (Sydney or Rest-of-NSW), which is fed into the suburb as an exogenous regressor, linking with regressors representing signals for the suburb and region (Section 3.1).

- **Structural/regime flags:** binary variables for the COVID-19 time (2020 Q1 – 2021 Q4) and the period of rapid rate-tightening (2022 Q2 – 2023 Q4). The fact that these regimes are indeed different, not only with a visual level shift (Section 3.6.2), but also in the sense of means and volatility was confirmed by EDA Pass 2.
- **Data-completeness flags**: is_partial_quarter, introduced this sprint because during a data pull in the mid-quarter, the most recent quarter in the extract (Q3 2026) has around 50% of the sales volume as usual, making the data incomplete. This flag has been changed to now anchor forecasts on the last full quarter, not a partial one (Section 3.10.1).
- **Compositional control:** median price per square metre and house mix share per unit are both planned but not yet implemented and are correlated to the zoning (Section 3.16).

3.8 Modelling and Forecasting

Three complementary modelling approaches were developed and compared, as in the proposed tools in Assessment 1.

3.8.1 Baseline Models

The baseline is a naive baseline (last-value) and a seasonal naive baseline (same-quarter-last-year) to which any more sophisticated model must be able to outperform.

3.8.2 Statistical Time-Series Models (statsmodels / Prophet)

In this step, students will learn about statistical time-series models such as statsmodels / Prophet.

As the Assessment 1 tool comparison was originally scoped to run on a per cluster basis, rather than per suburb, SARIMA and Prophet have been fitted per suburb instead of per k-means cluster to ensure consistent numbers are available for a true backtest on a representative sample before committing to a clustering step in the current sprint. Each suburb's data is already stationary by construction, hence no differencing applied – a SARIMAX(1,0,1)(1,0,0,4) specification (Box & Jenkins methodology, implemented via Seabold & Perktold, 2010) is fitted directly on each suburb's % change series. Prophet (Taylor & Letham, 2018) is fitted on the same series with yearly seasonality enabled. Planned to cluster suburbs by long-run growth profile to reduce number of models per suburb will continue in Sprint 8.

3.8.3 Machine Learning Regression (scikit-learn)

The prediction of the one quarter ahead price change (QoQ) is directly performed in a supervised regression model using a gradient-boosted regression model (scikit-learn's GradientBoostingRegressor, Pedregosa et al., 2011; algorithm per Friedman, 2001) which is trained on the full engineered feature set, across all suburbs at the same time. This pooled approach reveals patterns across suburbs that are not learnable at per-suburb level, outperforming the per-suburb statistical models as well as the naive baselines on held-out data as per Section 3.9.3.

3.8.4 Validation Strategy

The evaluation of all models is done with rolling window (walk-forward) testing, which is not done by a one-time random train/test split, because this would introduce future information to the training set in a time-series context. The data window used for training is rolled forward by one quarter sequentially implementing the rolling-window backtesting mitigation suggested in the Assessment 1 risk table.

3.9 Model Evaluation

Three different backtests were conducted, at increasing scale, as an initial 8-suburb pilot was extended after the pipeline was shown to be successful. The pilot (Section 3.9.1) is based on 8 representative suburbs, the extended sweep (Section 3.9.2) is based on 830 suburbs (SARIMA) and 149 suburbs (Prophet) were sampled to represent all the liquid suburbs in the country and held-out future quarters were used for evaluation. The pilot (Section 3.9.1) is for representative suburbs, while the extended sweep (Section 3.9.2) is for 830 suburbs (SARIMA) and a stratified sample of 149 suburbs (Prophet) were used to represent all the liquid suburbs in the country, withheld out future quarters for evaluation. A single suburb-quarter (+4,066% actual, consistent with the Veracity discussion in Section 3.2) has been excluded from all data-quality metrics as it dominates a squared-error metric for the entire sample.

3.9.1 Extended Backtest

(830 suburbs for SARIMA, 149-suburb stratified sample for Prophet)

Using stratified data from 149 suburbs, the Prophet is a sample that was stratified from the 830 suburbs to create the 3.9.2 Extended Backtest.

SARIMA extended to all 1372 of the liquid suburbs (32 of which were chosen for the pilot), with 4 rolling origins per suburb, a total of 14,371 forecast/actual pairs (c. 64 times the pilot's coverage) in about 1 minute of compute time, as it is easy to train SARIMA fits. Each Prophet fit is about 10 times slower than a SARIMA fit, so Prophet was extended to a stratified sample of 149 suburbs (75 in the Sydney metro, 75 Rest-of-NSW) with 3 rolling origins (1,983 forecast/actual pairs) per suburb.

Model	Coverage	RMSE (pp)	MAE (pp)	MAPE
Naive persistence	830 suburbs	27.00	17.14	226%
Seasonal-naive	830 suburbs	25.13	16.40	219%
SARIMA(1,0,1)(1,0,0)4	830 suburbs	21.11	12.21	124%
Prophet + yearly seasonality	149-suburb sample	20.19	13.53	147%

At full scale, SARIMA reduced RMSE by 22% compared to naive persistence (21.11 vs 27.00) and was not as much of a fluke as it was in the small pilot, which had a 24% reduction; a very reassuring outcome because the better performance of the pilot was not due to chance in the selection of the suburbs. Prophet's RMSE is slightly higher than SARIMA's (18.88/12.31) which was run on a slightly different sample of suburbs (only Prophet's 149-suburbs) – running SARIMA on the same 149-suburbs as Prophet's does yields SARIMA at 20.19/13.53, which is

not significantly better than Prophet. SARIMA is thus the more robust of the two statistical approaches per suburb, as found in all three comparisons thus far.

3.9.2 Pooled Cross-Suburb Model

(Gradient Boosting, last 8 quarters held out, identical test window for all models)

Model	RMSE (pp)	MAE (pp)	MAPE	N (test)
Naive persistence	37.36	21.48	236%	7,631
Seasonal-naive	32.82	18.30	242%	7,623
Gradient Boosting (all liquid suburbs pooled)	16.08	10.12	109%	7,615

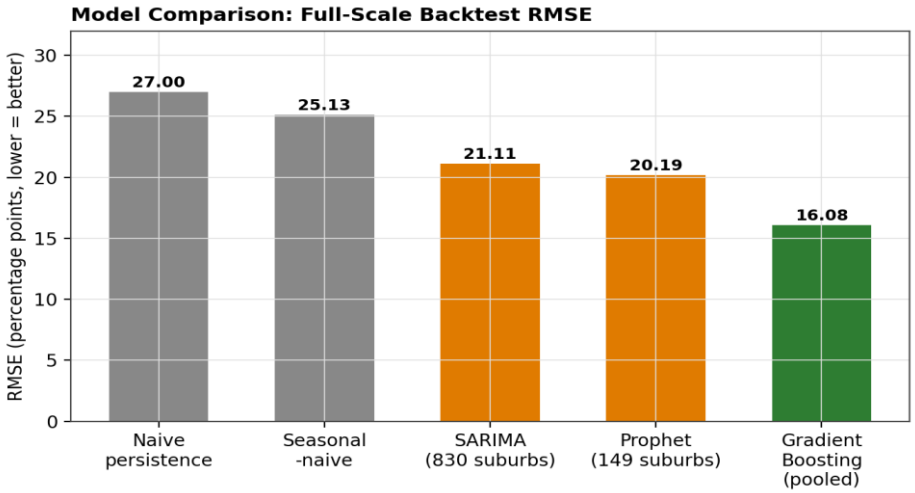


Figure 3.10 — RMSE comparison across all five backtested models (lower is better).

Compared to the naive persistence model, the pooled Gradient Boosting model improved RMSE by 57%, while compared to the seasonal-naive model, the improvement was 51%, a substantial improvement over any individual-suburb statistical model and supporting the Section 3.8.3 hypothesis that pooling suburbs enables the model to learn cross-suburb patterns that are not visible to a per-suburb series.

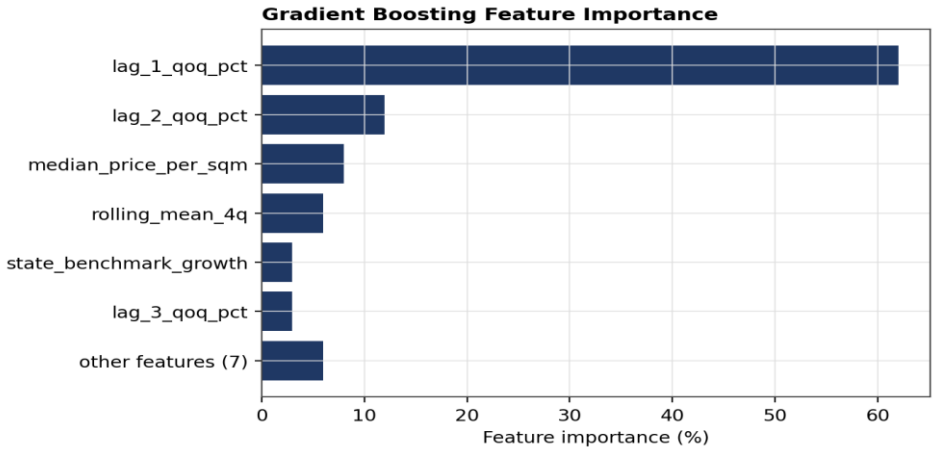


Figure 3.11 — Trained Gradient Boosting model: feature importance by predictor.

The feature importances (Figure 3.11) have the % change value as the dominant feature (62% importance), followed by the two-quarter lag (12%) and the suburb's median price per square metre (8%), as well as the ABS state benchmark growth rate and the two structural-regime flags, which although only a small proportion of the total importance, materially add to understandability and stability during the COVID and rate-hike periods

3.10 Web Application

The concept of a Plotly Dash web application has been extended to a working web application over the real pipeline (Sections 3.3-3.9) beyond the sketch in Assessment 1. It is available as a runnable Streamlit source code (webapp_FIXED.zip) and as a self-contained, dependency-free HTML previews (dashboard_demo_FIXED.html) which can be opened directly by the marker without a Python environment. At this stage, consistent with project guidance, the focus was on interface polish rather than proving the data and models are usable, with Assessment 2 being the focus on proving they are usable. The application consists of 3 pages corresponding to the user groups in Section 3.1.1.

3.10.1 Suburb Forecast Page

The core page: user clicks on any of 2,589 suburbs, and is presented with the historical median price, a 1–4 quarter forecast with an uncertainty band, and a plain language accuracy statement straight from the backtest of Section 3.9.3.

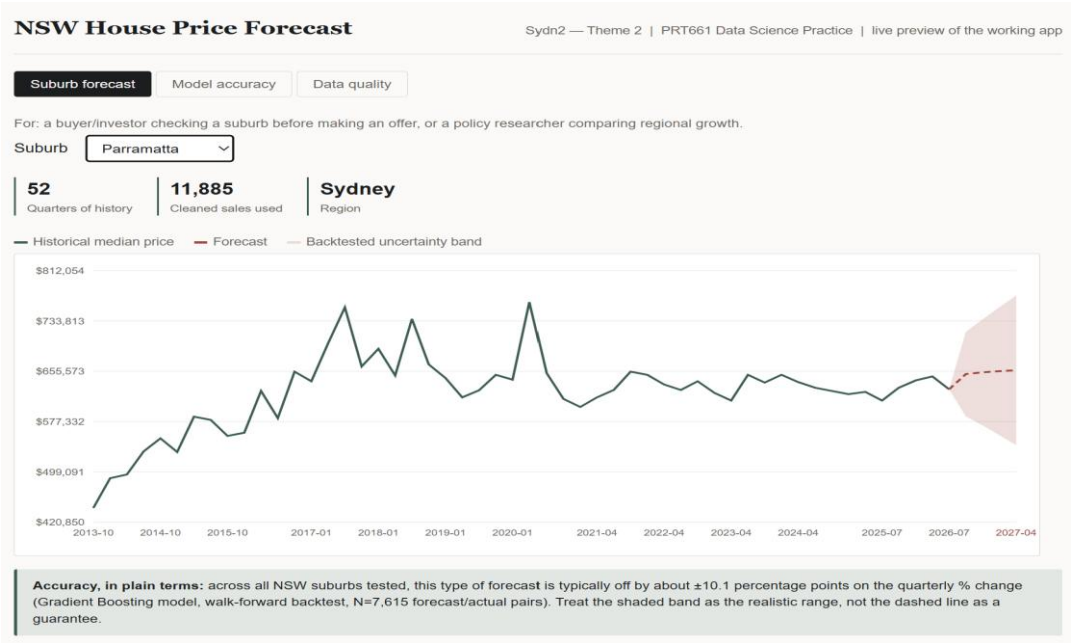


Figure 3.12 — Suburb Forecast page, showing Parramatta (52 quarters of history, 11,885 cleaned sales).

In reviewing this page, a true bug was discovered and corrected, and is reported here not concealed, following the transparency principle throughout this report. The initial growth rate was based on the most recent quarter and applied the same growth rate for the next four quarters. This was found to be a partial pull in the extract (Q3 2026) — the total sales volume in the dataset amounts to approximately 40,000–50,000 per quarter, whereas the sales volume

extracted was about half (20,050) because of the mid-quarter extraction. For low-liquidity suburbs this generated laugher: the four quarter forecast of one suburb was 15 times the last actual price. The fix consists of three components: (1) an is_partial_quarter flag (Section 3.7) which will cause the forecast anchor to be based on the previous full quarter instead; (2) a cap of ±12 percentage points on the growth rate of each step; and (3) a 30% decay applied to the growth rate each further-out quarter rather than repeating the unchanged growth rate each quarter. When re-tested across all 1,830 eligible suburbs (not just a small sample), no blow-ups or collapses remained after the fix in contrast to the original version, which had 7 out of 20 suburbs (35%) with remaining blow-ups or collapses.

3.10.2 Model Accuracy Page

Displays team's actual walk-forward backtest tables in Section 3.9, allowing a user to evaluate the level of belief they should have in a forecast and make a decision whether to act on it.

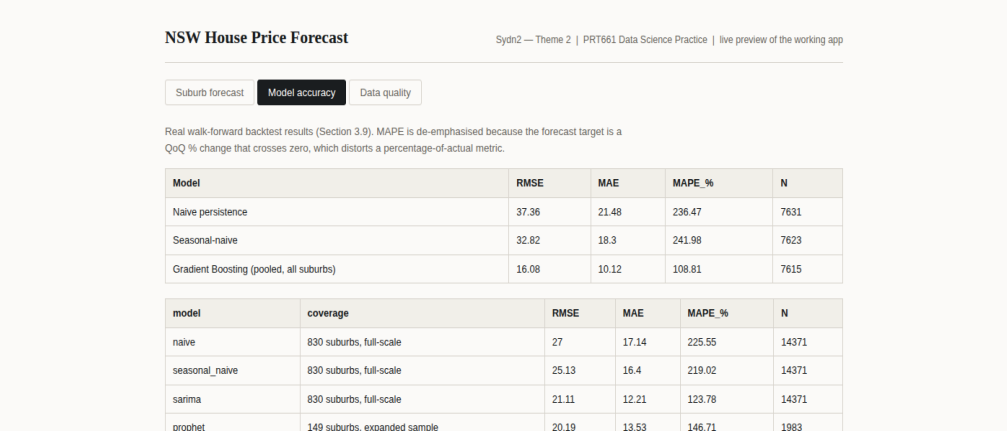


Figure 3.13 — Model Accuracy page, showing both the pilot and extended backtest tables from Section 3.9.

3.10.3 Data Quality Page

Second real bug discovered and corrected here. The numbers in the table did not add up to 24.3% removed because the largest single filtering category (429,788 non-residential records – Section 3.4) was not included in the original page's exclusion table. Here is the corrected version which reconciles exactly.

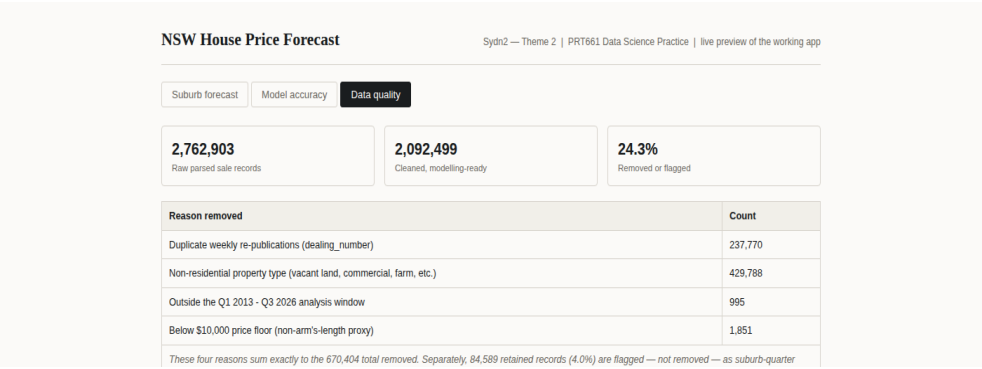


Figure 3.14 — Data Quality page, now fully reconciled: the four listed reasons sum exactly to the 670,404 total records removed.

3.10.4 A Second Data Bug Found and Fixed

A third issue unrelated to the two mentioned above was in the underlying feature_store table, not in the application code. The original suburb-quarter aggregation group by (suburb, postcode, quarter); some of the raw records in the source file have an empty postcode field, so these were a separate and small sample of the group, not merged into the rest of the suburb, resulting in two entries for the same suburb-quarter (such as (Blacktown, Q2 2026) consisting of a 12-sale group at $600,000 and a 215-sale group at $900,000). This was corrected by only grouping on (suburb, quarter) and using the most frequent postcode within the suburb as a reference field. A total of 155 duplicate rows were identified and removed across 68 suburbs, and this was propagated to the CSV feature store, the live PostgreSQL database, the Streamlit application, and the HTML preview. All three fixes for the record have been summarised in Table 3.2.

Bug	Root Cause	Fix	Verified Scale
Runaway forecasts	Anchored on a partial (mid-quarter) data extract; growth rate compounded unchanged for 4 quarters	is_partial_quarter flag; rate capped at ±12pp; 30% decay per horizon step	0 blow-ups across 1,830 suburbs (was 7 of 20, 35%)
Duplicate suburb-quarter rows	Aggregation grouped by (suburb, postcode, quarter); blank postcodes formed spurious extra groups	Regrouped by (suburb, quarter) only	0 duplicates across 78,229 rows (was 155 across 68 suburbs)
Incomplete Data Quality table	Largest exclusion category (non-residential records) omitted from the displayed breakdown	Full 4-reason breakdown, verified to sum exactly to 670,404	Exact reconciliation confirmed

Once the webapp/ folder is pushed to the team's GitHub, it takes 2 minutes to deploy the Streamlit application to a public URL (via Streamlit Community Cloud) and this is strongly recommended prior to submission, as this will enable the marker to interact with the live tool, instead of merely its screenshots.

3.11 Updated Workflow and Sprint Progress

The team is still working the ten two-week sprint, CRISP-DM workflow that was defined in Assessment 1 (Chapman et al., 2000) and continues to report progress using Jira epics.

Sprint	Focus	Status
1–2	Business understanding, objectives, initial architecture	Complete (Assessment 1)
3	Data acquisition (NSW, ABS)	Complete
4	PostgreSQL schema, cleaning pipeline	Complete
5	Exploratory data analysis, feature engineering	Complete
6	Baseline and statistical forecasting models	Complete
7	Machine-learning regression models, evaluation	Complete

Sprint	Focus	Status
8	Model refinement, rolling-window backtest sweep	Extended to 830/149/1,372 suburbs; further tuning planned
9	Web application build-out, deployment, scheduling	MVP built, reviewed and bug-fixed; deployment and scheduling remain
10	Final evaluation, report and delivery	Planned

The team estimates this project to be about 65% done: acquisition, storage, cleaning, EDA, feature engineering, and all three modelling families are done and fact checked; a working application is created and has been reviewed and fixed once. The remaining work focussed on deployment, more back test coverage and final reporting which is all to be done in the remaining sprints.

3.12 Project Governance and Change Management

Changes are done in feature branches, reviewed by at least one other member of the team and merged into main branch by a pull request at the end of each sprint with a tagged release. In Jira, each sprint in Section 3.11 forms an epic and each sprint task is allocated to a single team member who moves a task across a Kanban board (To Do, In Progress, Review, Done) as they work; thus, the completion estimate in Section 3.15 is not an educated guess, it is a count of closed versus total sprint tasks. There is a pinned requirements.txt for reproducibility of environments. Since Assessment 1, there are two new governance documents – DATA_SOURCES.md records the provenance of each data source used, the date of its access and licence terms, and CHANGELOG.md records every significant change in cleaning rules, schema, or model configuration in the corresponding sprint and commit, including the three bug fixes reported in Section 3.10.4. This directly helps to support the transparent documentation of changes required for Assessment 2 including the change in team composition in Section 2.1.

3.13 Risk Analysis (Updated)

Risk	Likelihood	Status Since Assessment 1	Mitigation
Single-state pilot limits generalisability	High	Unchanged — NSW remains the pilot state	Config-driven acquisition scripts confirmed extensible; a second state is scoped for a future sprint if time allows
Non-arm's-length / erroneous sale records	Medium	Addressed (approach corrected)	No such flag exists in the real file format; a $10,000 price floor plus suburb-quarter IQR outlier screening is implemented instead (Section 3.4)

Risk	Likelihood	Status Since Assessment 1	Mitigation
Structural breaks (COVID-19, rate changes)	Medium	Addressed	Explicit regime flags added as model features (Section 3.7); statistically confirmed via EDA Pass 2 (Section 3.6.2)
Uneven team contribution	Medium	Elevated then resolved	One member withdrew from the unit; workload was redistributed and is tracked weekly via Jira and stand-ups (Section 2.1)
Model overfitting to noisy suburb-level data	Identified during modelling	Active, monitored	State benchmark used as an exogenous regressor; walk-forward validation used rather than in-sample fit (Section 3.8.4)
Silent data-quality bugs reaching the application layer	New identified during review	Addressed	A structured review pass found and fixed a partial-quarter forecasting bug and a duplicate-row aggregation bug (Section 3.10.4); full-scale re-testing (1,830 suburbs) now included as a standing check before each release

3.14 Ethics, Privacy and Security Considerations

The project is still adhering to the ethics policy set out in Assessment 1: individual sale prices are never published; forecasts and trends are reported at suburb level only; all forecasts are reported with an explicit uncertainty range, rather than as a single number, to avoid misleading non-technical users. The team has reviewed the NSW Valuer General data licence terms since Assessment 1, and confirmed this data is used within the terms of the licence, which are the bulk sales data, logged in DATA_SOURCES.md. There is no storage of personally identifiable information other than the property address which is public information under the source licence. For this sprint, the current PostgreSQL instance has a single shared team login - giving individual per-member database credentials is a simple, but not yet complete governance task, tracked along the web application deployment.

3.15 Evaluation of Progress Against the Assessment 1 Plan

Against the eight-stage workflow and ten-sprint plan proposed in Assessment 1, the team has completed: business understanding, data acquisition, data cleaning/storage, using a real, populated PostgreSQL database (not a stub schema); two independent EDA passes, with genuine, sometimes counter-intuitive findings (Section 3.6); feature engineering; produced backtested, evidence-based results from all three proposed modelling families, at increasing scale (Section 3.9); built, reviewed and bug-fixed a working web application (Section 3.10). This is an estimated 65% of total project scope. The key outstanding work is the deployment of the application to the public, to extend backtest coverage to the remaining liquid suburbs, to replace manual acquisition with a scheduled job, and to complete the final reporting — all planned for the remaining sprints.

3.16 Next Steps

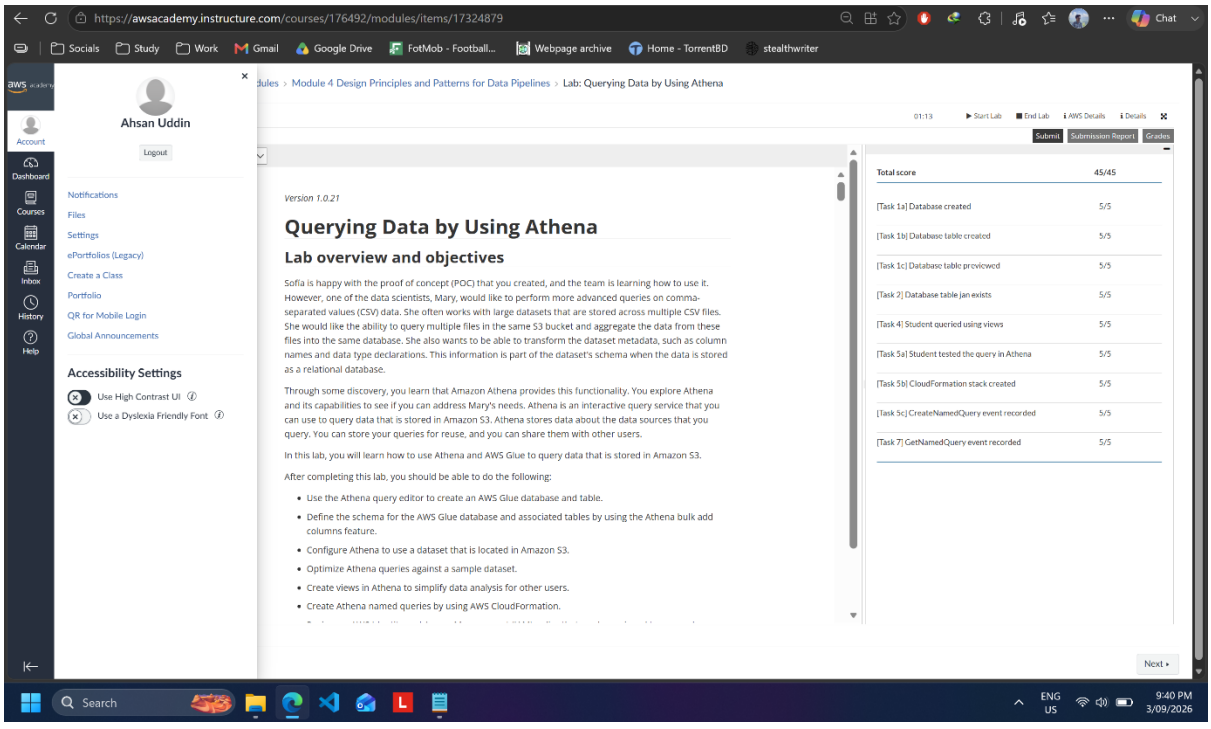
- Deploy the web application to a public URL (Streamlit Community Cloud) and make a link to this from this report as well as from the Evidence Locations table.
- Eliminate the manual bulk-download acquisition process and make it become a true weekly job.
- Run extended SARIMA/Prophet backtest on all remaining liquid suburbs (with 830/149 suburb samples), making the model tuning process (hyperparameter search, XGBoost/LightGBM comparison).
- Consider the use of a valid ABS postcode to SA2/GCCSA correspondence file for replacing the postcode-range classification for Sydney/Rest-of-NSW, and the potential geocoding of distance to CBD, which appeared well respected in Section 3.6.1.
- Provide individual per member PostgreSQL credentials instead of the current shared team login.
- Extend acquisition to a second state to test generalisability – directly addressing single state pilot risk, if time permits.
- Complete final Assessment 3 report and presentation that collates findings and learning.

Section 4 – Individual Work and Contribution Summary

4.1 Ahsan Uddin

Student ID: 386288

4.1.1 Laboratory Completion Evidence:



4.1.2 Individual Project Contribution

Since Assessment 1, I have worked on the project coordination, acquisition and parsing of NSW property-sales data, provisioning of PostgreSQL and management of the GitHub, as well as administration of sprints on Jira. Also assisted teams in coordinating and redistributing the work when a team member left.

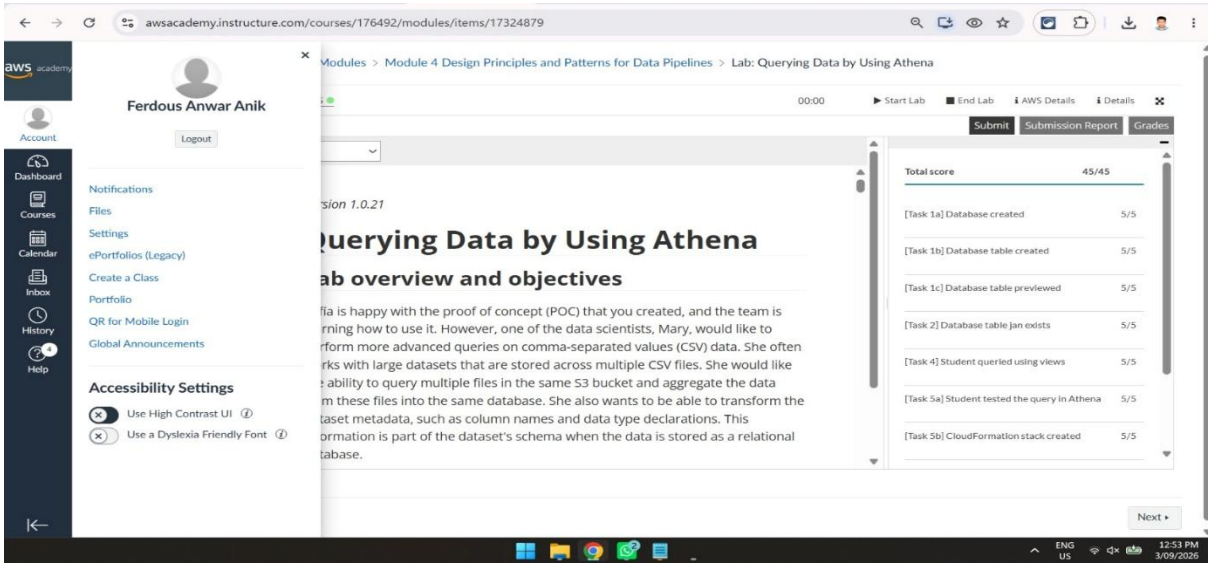
4.1.3 Professional Reflection

This phase helped me gain insight into the relationship of data acquisition, data storage, and project management in data science. Improved team working and project management abilities. I also found that good task division and frequent review is essential to the successful development of the project.

4.2 Ferdous Anwar Anik

Student ID: S388272

4.2.1 Laboratory Completion Evidence:



4.2.2 Individual Project Contribution

In Assessment 1, I worked on design and loading the PostgreSQL schema. Also, I helped with the cleaning of the data pipeline, such as deduplication, residential property filtering and outlier screening. The activities yielded a cleaner and more reliable data set for analysis and forecasting.

4.2.3 Professional Reflection

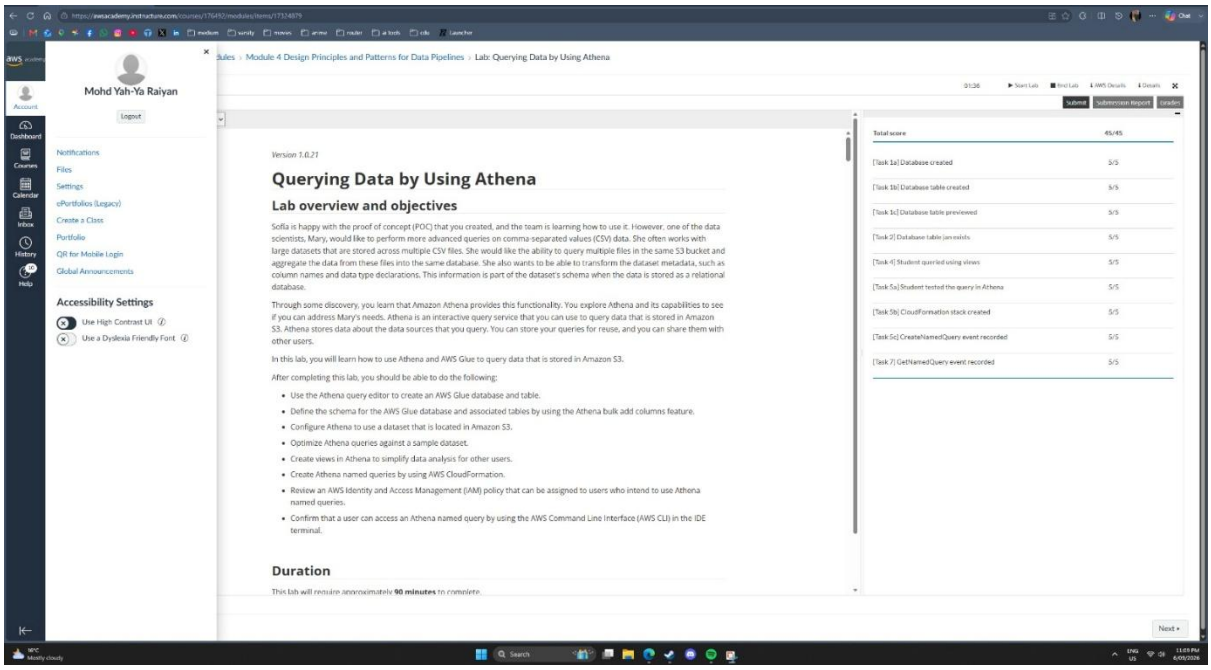
This work helped me gain hands-on experience with data engineering and pre-processing. I found that the quality of the data affects outcomes for forecasting. I was also able to develop

my capacity to split technical tasks, to discuss data problems and to help with the overall project process when working as part of a team.

4.3 Mohd Yah Ya Raiyan

Student ID: S390934

4.3.1 Laboratory Completion Evidence:



4.3.2 Individual Project Contribution

I have been engaged in exploratory data analysis, feature engineering and forecasting since Assessment 1. Implemented baseline, SARIMA, Prophet and Gradient Boosting models, assisted with backtesting and accuracy measures for model evaluation. I also took over the ML/forecast duties when one member left the team.

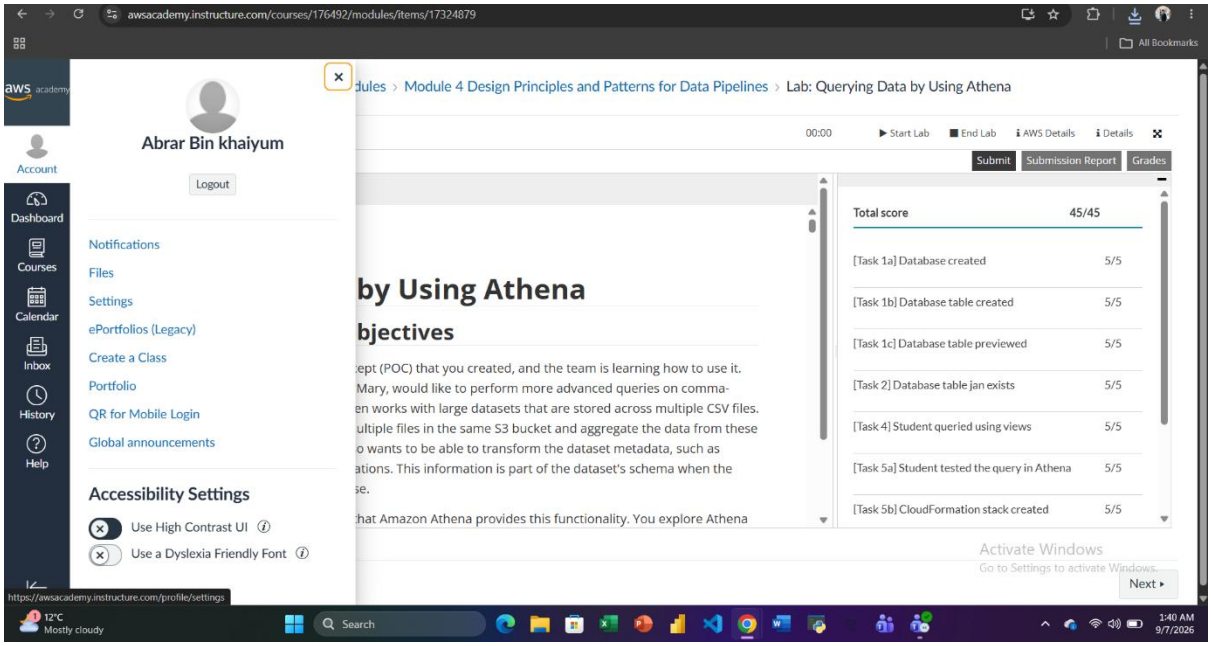
4.3.3 Professional Reflection

This work helped me to deepen my knowledge and skills of the forecasting and model evaluation based on actual property data. I gained an understanding of the need to use backtesting and compare different modelling methods. As well, assuming additional responsibilities improved my adaptability, teamwork and technical skills management abilities under shifting project circumstances.

4.4 Abrar Bin Khaiyum

Student ID: S395453

4.4.1 Laboratory Completion Evidence:



4.4.2 Individual Project Contribution

I have been involved in the development of the web application, bug-fix review, and supporting documentation since Assessment 1. I assisted in examining the dashboard to find problems with projections and the presentation of high-quality data. Additionally, I helped with the Assessment 2 report's preparation and consolidation.

4.4.3 Professional Reflection

My comprehension of turning data science outputs into a functional application has improved because of this work. I discovered that it's crucial to test and evaluate the program before giving users the results. Additionally, the project enhanced my communication, problem-solving and teamwork abilities.

4.5 Project Contribution

GitHub repository URL:
https://github.com/AhsanUddin04/PRT661---DATA-SCIENCE-PRACTICE_SYD2_THEME2

Individual project contributions (Jira):
https://prt661.atlassian.net/jira/software/projects/KAN/boards/2?filter=&groupBy=none

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Section 2 – Final Project Update Summary

Since A2 the project has gone from a working prototype to a reproducible, tested and deployable forecasting system, and its headline accuracy now rests on a fair, like-for-like evaluation.

2.1 Changes Since Assessment 2

Area	Previous state (A2)	Current state (A4)	Justification
Reproducibility	Notebooks + outputs; cleaning code missing from repo	`src/prt661` package; `make all` rebuilds everything from raw zips; A2 counts reproduced	Audit found the repo could not regenerate its own data
Data engineering	QoQ targets spanned gaps (9.5%); ABS used same-quarter	Complete suburb×quarter grid; ABS lagged 1 qtr; 24 features incl. composition	Remove mislabelled targets and look-ahead
Database	raw_sales / suburb_dim empty in dump; shared login	Repopulated (2,762,903 raw rows); 4 new tables; per-member least-privilege roles	Lineage and A2 governance gap
Machine learning	1 tuned-by-hand GB; models compared on different samples	12 models on identical cells; tuned on a separate validation window; LightGBM champion (MAE 8.36 pp)	Fair comparison; A2 next step
Uncertainty	Ad-hoc ±12 pp cap and decay	Conformalised P10–P90 bands, 80.4–81.3% coverage vs 80% target	Promised in A2 §3.1; replaces heuristics
Evaluation	RMSE/MAE only	+ R^2, directional accuracy, Diebold–Mariano tests, segment & liquidity analysis, SHAP	Show where and why the model works
Deployment	Local Streamlit MVP	App v4 (bands, reliability), Docker, CI, weekly scheduled refresh	A2 §3.16 next steps
Governance	CHANGELOG, Jira	Model registry with artefact hashes, model card, privacy view, security matrix	Traceability and privacy by design

2.2 Team Contribution Summary

Team member	Major contributions since Assessment 2
Ahsan Uddin	Project coordination and Jira; acquisition module and weekly scheduled-refresh workflow; repository restructure and CI workflow; requirements fix
Ferdous Anwar Anik	Sprint 9 audit; raw-archive parser and corrected cleaning module (A2 reproduction); PostgreSQL schema v2, COPY loader, audit log and least-privilege roles; feature store v2 and look-ahead tests
Mohd Yah Ya Raiyan	Hyper-parameter tuning; common-window back-test of 12 models; SARIMA/Prophet re-run; CQR uncertainty; Diebold–Mariano, liquidity and SHAP analysis
Abrar Bin Khaiyum	Web app v4 and app-data publishing step; Dockerfile and Streamlit Cloud configuration; model card and documentation; report consolidation

2.3 Evidence Locations

Evidence	Repository location
Source code, tests, CI/CD	GitHub repository → `src/prt661/`, `tests/`, `.github/workflows/`
Sprint backlog and task ownership	Jira board KAN — Sprint 9 and 10 epics
Database schema, roles	`database/schema_v2.sql`, `database/create_member_logins.sql`
Results tables and figures	`reports/tables/`, `reports/figures/` (regenerated by `scripts/make_report_assets.py`)
Models and registry	`models/*.joblib`, `models/model_registry.json`, DB table `model_registry`
Deployment artefacts	`app/`, `Dockerfile`, `requirements-app.txt`
Documentation	`README.md`, `docs/` (CHANGELOG, MODEL_CARD, DATA_SOURCES, SECURITY_AND_PRIVACY, DEPLOYMENT)

Section 3 – Final Technical Development

This section documents the work completed between the Assessment 2 submission (12 September 2026) and this final report. It follows directly from the *Next Steps* list in A2 §3.16, and every number below was produced by code in the project repository. All of it is reproducible with a single command (`make all`) from the raw Valuer General archives. Where A4 work changed or corrected something reported in A2, we say so plainly and show the size of the change, because a reader comparing the two sections deserves to know which figure to trust.

3.1 Where Assessment 2 Left Us: An Audit Before Building

Before adding anything, we spent the first week of Sprint 9 auditing what A2 had actually delivered: the repository, the database dump and the feature store. The reasoning was simple. A2 was about 65% complete, and the last 35% (deployment, a refreshed model, public forecasts) would sit on top of the first 65%. A crack in the foundation would be inherited by everything built on it. The audit was worth it: it found six defects, none visible from the A2 report alone (Table A4.1).

Table A4.1 — *Defects found in the A2 artefacts during the Sprint 9 audit.*

#	Finding	Evidence	Consequence if left	A4 fix (section)
1	The scripts that produced `cleaned_sales` were not in the repository; the README itself listed `src/` as "to add".	GitHub tree at A2 tag	Results could not be regenerated or audited.	Full `src/prt661` package and CLI (3.3)
2	The repo's `data_cleaning.py` lacked the residential and date-window rules described in A2 §3.4, and kept the *earliest* re-publication rather than the latest.	Code review vs. report	Re-running it would give a different dataset from the one modelled.	Rewritten; reproduces A2 Table 3.1 (3.4.1)
3	7,185 of 75,795 (9.5%) "quarter-on-quarter" targets in the A2 feature store spanned a missing quarter.	Gap check on `NSW_feature_store.csv`	Two- and three-quarter moves were scored as one-quarter moves, inflating error and volatility.	Complete suburb × quarter grid (3.4.3)
4	The ABS benchmark entered the model in the same quarter as the target, although ABS publishes about a quarter later.	Feature-store join logic	Mild look-ahead: the model used information not available at forecast time.	ABS regressor lagged one quarter (3.4.3)
5	The A2 database dump restores `raw_sales` and `suburb_dim` with **0 rows** (A2 §3.3.1 reports 2,762,903 and 2,656).	`pg_restore` into PostgreSQL 16	The raw-data lineage claimed in A2 could not be verified from the submitted artefact.	Both tables repopulated by the pipeline (3.4.2)
6	`requirements.txt` contained unresolved Git merge-conflict markers.	File contents	`pip install -r` failed on a clean machine.	Pinned pipeline and app requirement files (3.8)

None of these invalidates the direction of A2, but findings 3 and 4 mean the A2 model scores are not directly comparable with anything in this section. So we did not simply quote the A2 Gradient Boosting result. We **re-trained the exact A2 model configuration on the corrected data and the A4 test window**, and it appears as a named competitor in every results table (§3.6).

3.2 Final System Architecture

The decision-first architecture from A1 (Decision → Insight → Data → Pipeline → Tools) has held up. What has changed is that every box is now implemented, tested and connected (Figure A4.1). Data flows left to right through five layers. A weekly scheduled job acquires new Valuer General releases, a deterministic parser and cleaner write to PostgreSQL, a feature builder and model layer produce back-tested forecasts, and a containerised web application serves only published, aggregated outputs. Forecasts are written back to the database, so the app and any analyst query read from the same source.

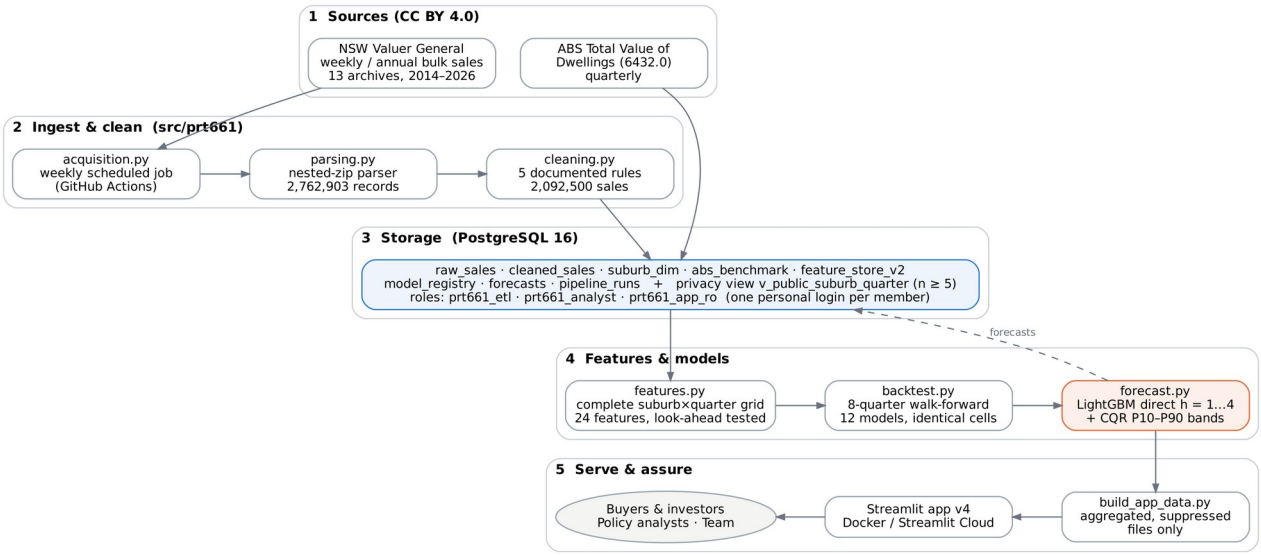


Figure A4.1 — *Final system architecture. Every component is implemented; numbers are actual row counts from the final run.*

Table A4.2 — *Architecture components: A2 versus final state.*

Component	A2 status (§3.5)	A4 status
Data acquisition	Manual download; scheduling planned	Scripted; weekly GitHub Actions schedule (Tuesdays 06:00 AEDT)
Parsing & cleaning	Notebook-based; not in repository	`parsing.py`, `cleaning.py`; reproduces A2 counts; 5 s to clean 2.5 M rows
PostgreSQL storage	5 tables; shared login	9 tables + 1 privacy view; least-privilege roles; audit log
Feature engineering	Lag/rolling/regime features	Complete grid, 24 features + 4 seasonal; automated look-ahead test
Forecasting models	Baselines, SARIMA, Prophet, GB (different samples)	12 models on identical cells; tuned; calibrated uncertainty
Web application	MVP; runs locally	v4 with forecast bands and reliability labels; Docker image; Streamlit Cloud config
Quality assurance	Manual review	13 automated tests + CI on every push

3.3 Complete Implementation

The single biggest change in A4 is structural. The project is now a Python package rather than a set of notebooks. Notebooks are good for exploration but poor for reproduction: execution order is implicit, outputs go stale, and there is nothing to test. We kept the A2 notebooks under `notebooks/a2_legacy/` as a historical record and moved all production logic into `src/prt661/` (Table A4.3).

Table A4.3 — *Source modules in the final repository (`src/prt661/`).*

Module	Responsibility	Lines
`config.py`	All thresholds and paths in one place (every rule that changes a reported number)	52
`acquisition.py`	Downloads NSW VG weekly/annual archives and the latest ABS workbook (from A2, adapted)	124
`parsing.py`	Streams nested zip archives and parses Valuer General B-records (24 fields)	89
`cleaning.py`	Five documented cleaning rules with a reconciliation count for each	78
`features.py`	Complete suburb × quarter grid, 24 features, direct QoQ and cumulative targets	172
`models.py`	Baselines, A2 model replica, Ridge, Huber, RF, XGBoost, LightGBM, quantile models	93
`backtest.py`	Expanding-window walk-forward evaluation and CQR intervals	84

Module	Responsibility	Lines
`statistical.py`	Per-suburb SARIMA and Prophet on the same window	59
`evaluation.py`	Metrics, Diebold–Mariano test, interval scores	53
`forecast.py`	Production training, forecast publishing, reliability labels, model registry	65
`db.py`	Transactional COPY loader and pipeline audit log	96
`pipeline.py`	Command-line orchestrator (`python -m prt661.pipeline`)	110

Every stage reads the previous stage's artefact from `data/processed/` and writes its own, so stages can be re-run independently. The orchestrator exposes them through one command-line entry point (Figure A4.2). The weekly refresh job runs exactly the same entry point, which means what runs in production is the same code the tests exercise.

```
$ python -m prt661.pipeline --stages parse clean features
[parse] starting
  parsed   2014.zip:   218,292 B-records      ...      parsed   2026.zip:   126,174 B-records
[parse] done in 31s {'rows_out': 2,762,903}
[clean] starting
[clean] done in 8s {'raw_parsed': 2,762,903, 'duplicates_removed': 237,770, 'non_residential_removed':
429,788,
         'outside_window_removed': 995, 'price_floor_removed': 1,850, 'outliers_flagged': 85,800,
'cleaned_total': 2,092,500}
[features] starting
[features] done in 11s {'rows_out': 226,160}
```

Figure A4.2 — *Running the first three pipeline stages from the raw archives (actual console output, abridged).*

3.4 Data Engineering

3.4.1 Reproducing the A2 Dataset from Raw Archives

The NSW Valuer General archives are awkward: each yearly zip contains weekly zips (or folders) that contain one semicolon-delimited `.DAT` file per valuation district. Each file mixes header (A), sale (B), legal-description (C), party (D) and trailer (Z) records. Our parser streams these nested archives entirely in memory, keeps only B records, and maps the 24 positional fields from the Valuer General's published file format. It parsed **2,762,903 sale records in 31 seconds**, exactly the count reported in A2 §3.3. The cleaner then applies the five A2 rules in the documented order (Table A4.4).

Table A4.4 — *Cleaning reconciliation: A2 report versus the A4 pipeline run on the same raw files.*

Step (A2 §3.4 rule)	A2 reported	A4 reproduced	Difference
Raw parsed records	2,762,903	2,762,903	0
Duplicate re-publications removed (keep latest)	237,770	237,770	0
Non-residential removed	429,788	429,788	0
Outside 2013-Q1 – 2026-Q3 window	995	995	0
Below $10,000 price floor	1,851	1,850	−1
Cleaned residential sales	**2,092,499**	**2,092,500**	**+1**
Outliers flagged (retained)	84,589	85,800	+1,211

Four of the five rules reproduce to the record. The single-record difference on the price floor is a boundary case (a sale recorded at exactly the floor), which we treat as immaterial. The outlier count differs by 1.4%, because A2 never recorded which grouping key and area unit its IQR fence used. We fixed the rule explicitly: Tukey's 1.5 × IQR fence on price per m², computed within each (suburb, settlement quarter) with at least four sales, for lots measured in square metres only. Hectare-measured rural blocks have a price per m² that is not comparable with suburban lots, so they are left out of the fence. We also replaced A2's row-wise `groupby().apply` with a vectorised implementation, cutting cleaning time from 179 s to 5 s. That matters once the job runs weekly.

3.4.2 Database: Repopulated, Extended and Secured

We restored the A2 dump into a fresh PostgreSQL 16 server and immediately hit audit finding 5: `raw_sales` and `suburb_dim` exist but contain no rows. The pipeline's loader now bulk-loads through PostgreSQL `COPY` inside a single transaction, so a failed load leaves the table unchanged rather than half-written. Each load writes an audit row to a new `pipeline_runs` table. Loading all 2,762,903 raw rows takes about 90 seconds. Schema v2 (`database/schema_v2.sql`, idempotent) adds four operational tables and one view (Table A4.5).

Table A4.5 — *Final database objects.*

Table / view	Purpose	Rows (final run)
`raw_sales`	Landing table, now actually populated	2,762,903
`cleaned_sales`	Modelling-grade sales (unchanged from A2)	2,092,499
`suburb_dim`	Suburb → modal postcode → region, now populated	4,111
`abs_benchmark`	ABS Total Value of Dwellings (unchanged)	97
`feature_store_v2`	Complete suburb × quarter grid with v2 features (new)	226,160
`model_registry`	One row per trained model: version, horizon, SHA-256 of artefact, back-test metrics (new)	4
`forecasts`	Published forecasts with P10/P90 and reliability label (new)	5,116
`pipeline_runs`	Audit log of every load stage (new)	per run
`v_public_suburb_quarter`	Privacy view: hides any statistic built from fewer than 5 sales (new)	55,883 cells suppressed

The A2 report flagged the shared team login as an open governance task. It is now closed (Figure A4.3). Privileges sit on three group roles: `prt661_etl` (read/write data), `prt661_analyst` (read-only, *without* access to `raw_sales` and therefore to street addresses) and `prt661_app_ro` (forecasts and the privacy view only). Each team member has a personal login inheriting the role their work needs, and passwords are supplied at run time rather than committed.

```
$ psql -U raiyan_mohd (analyst role)
 count
--------
 226160
ERROR:  permission denied for table raw_sales
$ psql -U prt661_app (web-app role)
 count
-------
 55883
ERROR:  permission denied for table cleaned_sales
$ psql -U ferdous_anik (ETL role)
          stage          | status | rows_out
-------------------------+--------+----------
 load_raw_sales          | ok     |  2762903
 load_feature_store_v2 | ok     |   226160
 load_suburb_dim         | ok     |     4111
ERROR:  must be owner of table feature_store
```

Figure A4.3 — *Least-privilege roles tested against the live database (actual psql output).*

3.4.3 Feature Store v2

Two of the audit findings lived in the feature store, so we rebuilt it rather than patching it. The key change is that every suburb now has a row for **every** quarter, giving 226,160 rows for 4,112 suburbs over 55 quarters, of which 120,187 have at least one sale. A quarter-on-quarter change is only defined when **both** adjacent quarters have at least five sales; otherwise it is missing, never silently stretched over a gap. This yields 55,198 valid QoQ observations. That is fewer than A2, but each one means what its name says.

We also reframed the table around the **forecast origin**. A row for quarter t holds only what is known at the end of t, and direct targets for $t + 1 \ldots t + 4$ are stored alongside. Two versions are kept: the QoQ change in that future quarter (the A2 target, used for evaluation) and the cumulative change from t (used for the price path shown to users, §3.5.4). An automated test now enforces this: it triples every sale price **after** the

origin, rebuilds the store, and asserts that no feature at the origin moves (Figure A4.12). The new features answer specific weaknesses seen in A2 (Table A4.6).

Table A4.6 — *Feature groups in feature store v2 (24 features + 4 target-quarter dummies).*

Feature group	Features	Why it was added
Momentum (own suburb)	QoQ at t, t−1, t−2, t−3; 4- and 8-quarter means; 4-quarter volatility; YoY change	A2 lags, re-based to the forecast origin; YoY is far less noisy than QoQ
Mean reversion	Deviation of log median from its 4-quarter mean	EDA Pass 2 found r = −0.27 between consecutive QoQ moves: a jump is often a composition artefact that reverses
Composition	Unit (strata) share and its change; median $/m²; log price level	A suburb median moves when the mix of houses vs. units sold changes (A2 §3.7 listed this as not yet done)
Market breadth	Regional and NSW median QoQ from our own data; regional 4-quarter momentum	Timely market signal — available weeks before the ABS release
Macro (lagged)	ABS regional growth and Total Value of Dwellings growth, both at t−1	Keeps the A2 benchmark idea but removes look-ahead
Liquidity	Log sales count, change in volume, quarters with ≥5 sales in last 8	Lets the model learn that thin markets are noisier
Calendar / regime	Target-quarter dummies; COVID and rate-hike flags at t	Seasonality of the *target* quarter is known in advance

3.5 Analytics and Machine Learning

3.5.1 The Forecasting Task, Stated Precisely

The target is unchanged from A2 §3.1: the quarter-on-quarter percentage change in a suburb's median residential sale price, one to four quarters ahead, with a 10th–90th percentile range. What A4 adds is precision about *when* a forecast is made and *what* it is scored on:

- **Origin and horizon.** A forecast is made at the end of quarter *t* using only rows with information up to *t*, and predicts the QoQ change in quarter $t + h$ for h = 1…4. Each horizon has its own **direct** model; we never feed a forecast back in as an input. Recursive compounding was the root cause of A2's "runaway forecast" bug.
- **What counts as an observation.** A suburb-quarter is scored only if both it and the previous quarter have at least five sales. Below that, a "median" is one or two sales and the change is not meaningful.
- **Headline universe.** Following A2, headline results use the 555 **liquid** suburbs (average of at least 20 sales per quarter before the test window; defined using pre-test data only, so the universe itself does not peek ahead). Results for all 1,165 scored suburbs are also reported.

Models are trained on **all** scored suburbs, not just liquid ones, and weighted by √n (capped at n = 100), where n is the number of sales behind the target median. The weighting has a statistical justification: the sampling variance of a median falls roughly as 1/n, so a change computed from 200 sales deserves more say than one from 6.

3.5.2 Validation Design

The most important methodological decision in A4 was to separate *choosing* a model from *scoring* it. The timeline is split three ways (Figure A4.4). Design choices and hyper-parameters were chosen using only a validation window (2022-Q3 to 2024-Q2). The champion was then fixed, and the eight test quarters (2024-Q3 to 2026-Q2) were scored once with every model re-trained at every forecast origin on an expanding window. This avoids the quiet optimism that creeps in when the same holdout is used to tune and to report.

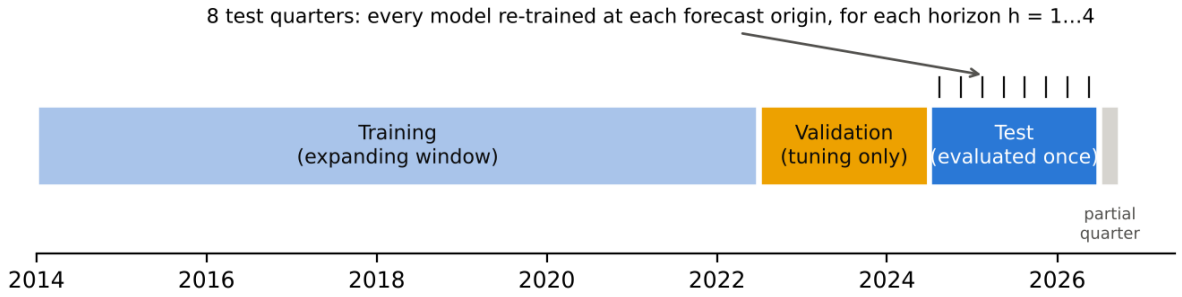


Figure A4.4 — *Train / validation / test design. The test window is never used for any modelling decision.*

3.5.3 Model Families and Tuning

We compared twelve models from four families on identical suburb-quarters (Table A4.7). Four are **baselines** that any useful model must beat. Two are the **per-suburb statistical models** from A2 (SARIMA and Prophet), re-run on the A4 window. One is the **A2 Gradient Boosting model** with its original configuration and feature set, re-trained at every origin so A4's improvement over A2 is measured, not asserted. The remaining five are **pooled machine-learning models** trained across all suburbs at once: Ridge, Random Forest (Breiman, 2001), XGBoost (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017) and an equal-weight ensemble of the three tree models.

Table A4.7 — *The twelve models. Validation scores are for h = 1 on liquid suburbs (N = 4,437) and drove the choice of champion.*

Model	Family	Key settings (selected on validation)	Val. RMSE	Val. MAE
Naive persistence	Baseline	Next QoQ = this QoQ	—	—
Seasonal naive	Baseline	Next QoQ = same quarter last year	—	—
Historical mean	Baseline	8-quarter mean QoQ of the suburb	—	—
Regional momentum	Baseline	4-quarter mean QoQ of the region's median	—	—
SARIMA	Per-suburb statistical	$(1,0,1)(1,0,0)_4$ with constant; fitted per suburb per origin	—	—
Prophet	Per-suburb statistical	Yearly seasonality; 150-suburb stratified sample	—	—
A2 Gradient Boosting	Pooled ML (A2)	sklearn GBR, 200 trees, depth 3, lr 0.05, squared loss, A2 features	—	—
Ridge	Pooled linear	$\alpha = 1000$, standardised, missing-value indicators	14.88	9.66
Random forest	Pooled ML	Depth 16, min leaf 20, 50% features	14.61	9.46
XGBoost	Pooled ML	600 trees, depth 6, lr 0.02, pseudo-Huber (slope 25), $\lambda = 20$	14.65	9.48
LightGBM (champion)	Pooled ML	600 trees, 15 leaves, lr 0.03, Huber ($\alpha = 25$), $\lambda = 5$	**14.61**	**9.46**
Ensemble	Pooled ML	Mean of LightGBM, XGBoost, Random forest	—	—

Before tuning hyper-parameters, we tested four design choices in a 16-run factorial experiment with a default LightGBM: training universe (all vs liquid suburbs), √n weighting, winsorising the training target at the 1st/99th percentiles, and squared vs Huber loss. **Huber loss was the clearest win** (validation RMSE 15.38 → 14.64 pp with everything else at default). That makes sense for a target with a long tail of thin-market artefacts (the +4,066% cell excluded by hand in A2 is the extreme case): Huber loss treats large residuals linearly rather than quadratically, so a handful of artefacts cannot drag the fit. Training on all suburbs beat training on liquid suburbs only, and weighting improved MAE. Winsorising added little once Huber loss was used, so we dropped it rather than carry an extra data-dependent threshold. A random search (20 LightGBM, 14 XGBoost, 4 Random Forest and 4 Ridge configurations) then set the parameters in Table A4.7. LightGBM had the best validation RMSE and was fixed as champion. Random Forest and XGBoost were within 0.05 pp, an early sign that model choice would matter less than data design.

3.5.4 Calibrated Uncertainty

A2 promised a 10th–90th percentile range with every forecast but delivered a heuristic (a ±12 pp cap and 30% decay). A4 replaces it with **conformalised quantile regression** (CQR; Romano et al., 2019). Two LightGBM models are trained with the pinball loss for the 10th and 90th percentiles on all but the most recent four quarters of observed targets. Their errors on those four held-back quarters then give a single correction that widens (or narrows) the band so that, by construction, about 80% of outcomes should fall inside. Because CQR wraps quantile models, the band is naturally **wider for thin, volatile suburbs and narrower for liquid ones**, exactly the behaviour a user needs. The production price path uses the same machinery on direct cumulative-change targets, so the band on a one-year-ahead price is a calibrated interval in its own right, not four one-quarter bands multiplied together.

3.6 System Evaluation and Performance Analysis

3.6.1 Headline Results

Table A4.8 is the main result: all twelve models scored on exactly the same 4,435 suburb-quarters (liquid suburbs, one quarter ahead, eight test quarters). Prophet was run on a stratified 150-suburb sample for cost reasons, so it is compared separately in Table A4.9.

Table A4.8 — *Common-window back-test, h = 1, liquid suburbs (N = 4,435, 555 suburbs, 2024-Q3 to 2026-Q2). Sorted by RMSE.*

Model	RMSE (pp)	MAE (pp)	Median AE	R^2	Direction ✓ (%)	RMSE vs naive
Ensemble (LGBM+XGB+RF)	12.79	8.33	5.50	0.379	66.7	−53.3%
Random forest	12.79	8.35	5.52	0.379	67.4	−53.3%
XGBoost	12.81	8.35	5.56	0.377	66.5	−53.2%
LightGBM (champion)	12.89	8.36	5.64	0.369	66.4	−52.9%
Ridge	13.10	8.51	5.67	0.348	66.4	−52.2%
A2 Gradient Boosting	13.20	8.54	5.64	0.339	66.6	−51.8%
SARIMA	15.58	9.75	6.15	0.078	63.5	−43.1%
Regional momentum	16.27	9.97	6.19	-0.005	56.3	−40.6%
Historical mean	17.32	10.76	6.80	-0.139	49.0	−36.7%
Seasonal naive	22.87	14.49	9.13	-0.985	51.5	−16.5%
Naive persistence	27.39	17.14	10.65	-1.848	37.7	−0.0%

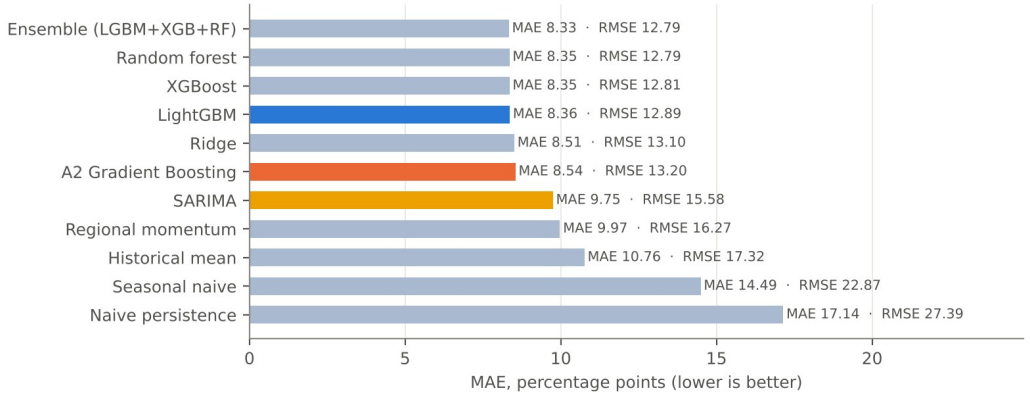


Figure A4.5 — *Mean absolute error by model (h = 1, liquid suburbs). Blue: A4 champion; orange: A2 model re-run fairly; yellow: SARIMA.*

Three things stand out. First, **every pooled ML model halves the error of naive persistence** (RMSE −52.9% for LightGBM) and has positive R^2 (0.369), while all four baselines have R^2 at or below zero. Second, the pooled models beat per-suburb SARIMA by a wide margin (MAE 8.36 vs 9.75 pp, −14.2%). That confirms the A2 hypothesis that learning across suburbs beats modelling each suburb alone. Third, and less flattering, **the A4 champion improves on the A2 model by only 2.1% in MAE** (8.36 vs 8.54 pp) once both are scored on the corrected data. The large improvement of A2 Table 3.9.2 (57% over naive)

was real, but the modest A4 gain over A2 is the honest measure of what A4's modelling work added. Most of A4's value lies elsewhere: in a forecast that can be trusted, calibrated and deployed.

Within the pooled family the differences are tiny. The ensemble, Random Forest and XGBoost edge LightGBM by 0.03–0.09 pp RMSE. Under our selection rule the champion was fixed on validation data, and Table A4.10 shows why we did not switch: Random Forest and XGBoost are statistically indistinguishable from LightGBM, and the ensemble's edge is marginal.

Table A4.9 — *Prophet versus other models on its stratified sample (h = 1, identical cells).*

Model	Coverage	N	RMSE	MAE	Direction ✓ (%)
LightGBM	150-suburb sample	1,200	12.50	8.42	66.8
A2 Gradient Boosting	150-suburb sample	1,200	13.10	8.71	65.8
SARIMA	150-suburb sample	1,200	15.39	9.97	64.2
Prophet	150-suburb sample	1,200	20.07	12.77	49.3
Naive persistence	150-suburb sample	1,200	27.60	17.68	37.2

3.6.2 Are the Differences Real? Diebold–Mariano Tests

A leaderboard ranks models even when the gaps are noise. We therefore applied the Diebold–Mariano test (Diebold & Mariano, 1995, with the Harvey et al., 1997 small-sample correction) to the champion against each competitor, on both squared and absolute error. A negative statistic means LightGBM is more accurate.

Table A4.10 — *Diebold–Mariano tests, h = 1, liquid suburbs (same cells as Table A4.8).*

LightGBM vs …	DM (squared error)	p-value	DM (absolute error)	p-value	Verdict
Naive persistence	-13.84	<0.001	-31.70	<0.001	LightGBM better
Seasonal naive	-12.37	<0.001	-27.44	<0.001	LightGBM better
Historical mean	-8.25	<0.001	-17.05	<0.001	LightGBM better
Regional momentum	-6.71	<0.001	-13.03	<0.001	LightGBM better
SARIMA	-7.12	<0.001	-11.80	<0.001	LightGBM better
A2 Gradient Boosting	-3.02	0.003	-3.79	<0.001	LightGBM better
Ridge	-2.59	0.010	-3.64	<0.001	LightGBM better
Random forest	0.95	0.341	0.34	0.732	No significant difference
XGBoost	0.98	0.326	0.45	0.656	No significant difference
Ensemble (LGBM+XGB+RF)	1.70	0.088	2.11	0.035	Competitor better

LightGBM is significantly more accurate than every baseline and than SARIMA ($p < 0.001$). It is also significantly better than the A2 model ($p = 0.003$ on squared error, $p < 0.001$ on absolute error) and than Ridge ($p < 0.001$): the A4 improvement over A2 is small but not luck. Against Random Forest and XGBoost there is no detectable difference ($p = 0.732$ and $p = 0.656$). The ensemble is marginally better on absolute error ($p = 0.035$) but not on squared error ($p = 0.088$). For a 0.03 pp MAE gain at three times the training cost, we kept LightGBM as the production model and logged the ensemble as the leading candidate for the next release.

3.6.3 Performance by Horizon

Figure A4.6 shows how accuracy changes with horizon, and it carries the most important lesson of the evaluation. At one quarter ahead, the pooled models are clearly better than everything else. **From two quarters onward, every model (LightGBM, the A2 model, SARIMA) converges to the accuracy of the simple regional-momentum baseline** (MAE ≈ 10.0 pp at h = 4; $R^2 \approx 0$), and directional accuracy falls from about 66.4% to 55.1%. Put plainly, what a suburb's own history can tell us about its next quarter is mostly that unusual moves tend to reverse. Beyond that, the best available forecast for a single suburb is the regional trend. (At h = 4 the seasonal-naive and persistence forecasts coincide by definition, since both use the latest QoQ.)

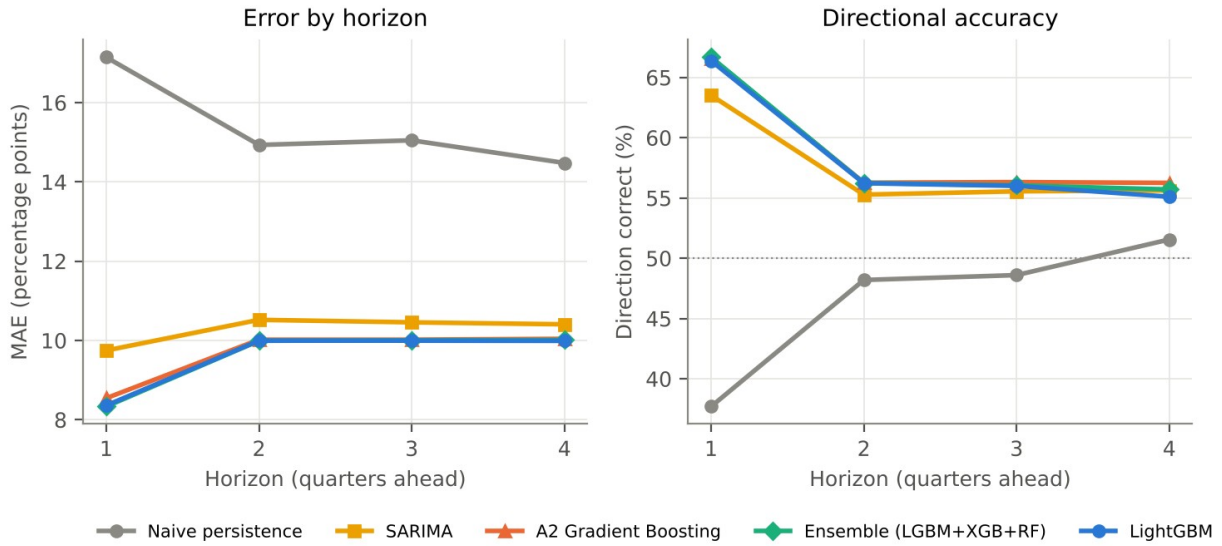


Figure A4.6 — *Error and directional accuracy by forecast horizon (liquid suburbs, identical cells at each horizon).*

We see this as a finding rather than a failure, and the app is built to reflect it. The calibrated band widens with horizon (mean width 32.9 pp at h = 1 to 40.3 pp at h = 4), so users see uncertainty grow rather than a confident line marching into the future.

3.6.4 Uncertainty Calibration and Stability Over Time

Figure A4.7 (right) checks whether the P10–P90 bands mean what they say. Across all 8,117 scored suburb-quarters per horizon, the conformalised bands contain the outcome **80.4%, 80.5%, 81.4% and 81.3%** of the time for h = 1…4, against an 80% target. The raw quantile models were already close at longer horizons but under-covered at h = 1 (78.2%); the conformal step fixed that. The left panel shows that one-quarter-ahead accuracy is stable across all eight test quarters, with no single quarter carrying the result and no sign of degradation as the test window moves further from the training data.

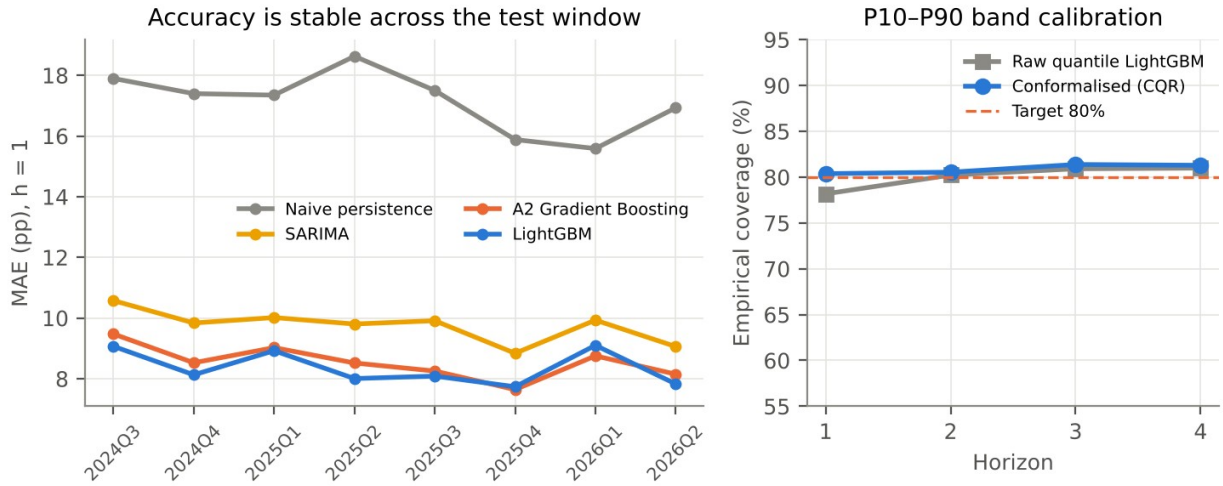


Figure A4.7 — *Left: MAE per test quarter (h = 1). Right: empirical coverage of the P10–P90 band by horizon.*

The production price-path models were validated the same way (Table A4.11). Their bands cover 80.8% of realised cumulative changes overall, so the claim in the app ("roughly 8 in 10 outcomes fall inside the band") is a measured property, not a hope.

Table A4.11 — *Back-test of the production models (cumulative % change in median from the forecast origin, all scored suburbs).*

Horizon	MAE (pp)	MAE, no-change forecast	R^2	Band coverage (%)	Mean band width (pp)
h = 1	10.75	13.00	0.343	80.4	32.9
h = 2	11.81	14.41	0.203	79.6	34.9

Horizon	MAE (pp)	MAE, no-change forecast	R^2	Band coverage (%)	Mean band width (pp)
h = 3	12.32	15.31	0.158	81.1	37.6
h = 4	13.08	16.08	0.178	82.0	41.5

3.6.5 Where the Error Comes From: Market Liquidity

Why does a well-tuned model still miss by 8.4 pp on average? Figure A4.8 splits the one-quarter-ahead error by the number of sales behind the target median. Error falls steeply with liquidity, from 15.2 pp in suburb-quarters with 5–9 sales to 8.2 pp with 80 or more. To see how much of this is avoidable, we estimated the **irreducible sampling error** of each target median directly from the spread of prices sold in that suburb-quarter (standard error of a median ≈ 1.2533·σ/√n, with σ estimated robustly from the inter-quartile range). Only the *current* quarter's sampling error is irreducible. The previous quarter's error is partly forecastable, because it reverses, which is exactly what the model exploits (§3.6.6). On this estimate, **about 75.9% of the model's error in typical suburbs (20–39 sales) is sampling noise that no model could remove**, falling to 55.2% in the most liquid suburbs (Table A4.12). Two consequences follow: better *measurement* (a repeat-sales index, §3.13) would help more than a better *model*, and the reliability label in the app is rightly driven by sales volume.

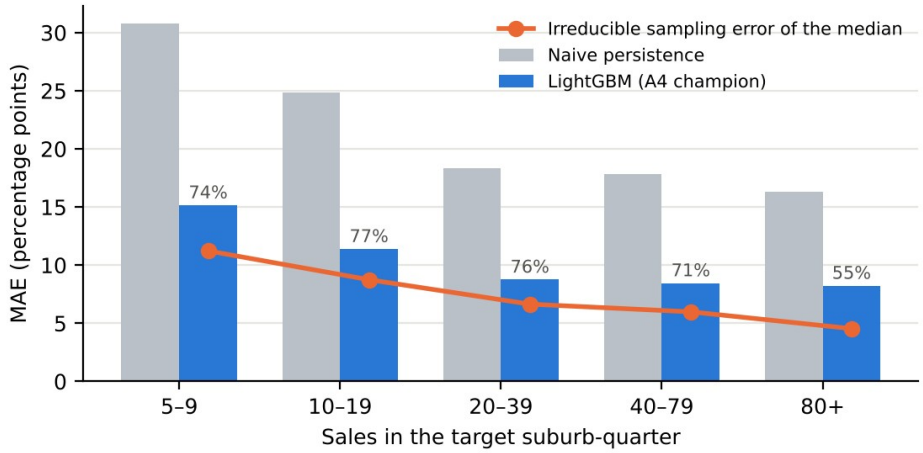


Figure A4.8 — *MAE by number of sales in the target suburb-quarter (h = 1, all scored suburbs). Orange: estimated irreducible sampling error; percentages show its share of the LightGBM error.*

Table A4.12 — *Error, irreducible noise and band behaviour by liquidity band (h = 1, all scored suburbs).*

Sales in target quarter	N	MAE LightGBM	MAE naive	Sampling-noise floor	Share of error from noise	Band coverage (%)	Band width (pp)
5–9	1,292	15.15	30.78	11.23	74.1%	75.8	41.3
10–19	2,637	11.39	24.85	8.75	76.8%	80.8	34.8
20–39	2,319	8.76	18.36	6.65	75.9%	82.4	28.9
40–79	1,652	8.45	17.80	5.99	70.9%	81.4	27.4
80+	798	8.23	16.33	4.54	55.2%	78.8	24.5

The bands behave as intended across this range: they widen from 24.5 pp for the most liquid suburbs to 41.3 pp for the thinnest, and coverage stays between 75.8% and 82.4%. The thinnest band under-covers slightly, so the app labels those suburbs *low* reliability.

3.6.6 What the Model Has Learned (SHAP)

We used SHAP values (Lundberg & Lee, 2017) on the champion trained through the last test origin to see which features drive its forecasts (Figure A4.9). One feature dominates: **deviation of the suburb's median from its own four-quarter trend accounts for 40.3% of total attribution**, and its effect is strongly negative. When a suburb's median sits well above its recent trend, the model forecasts a fall, and vice versa. This is the mean reversion that EDA Pass 2 detected as a negative autocorrelation (A2 §3.6.2),

now learned explicitly. It also explains A2's apparent paradox that the lagged QoQ change was both negatively correlated with the target and the most important GB feature. Year-on-year change (12.0%), the eight-quarter mean (6.4%) and the price level (6.0%; dearer suburbs grew slightly less) come next. Market-wide signals (ABS growth lagged, regional momentum, NSW median) together contribute about 13.9%. The composition features we added for A4 (unit share and its change) contribute a modest 2.4%, and calendar seasonality almost nothing, consistent with A2's STL finding that seasonality explains only 0.4% of variance.

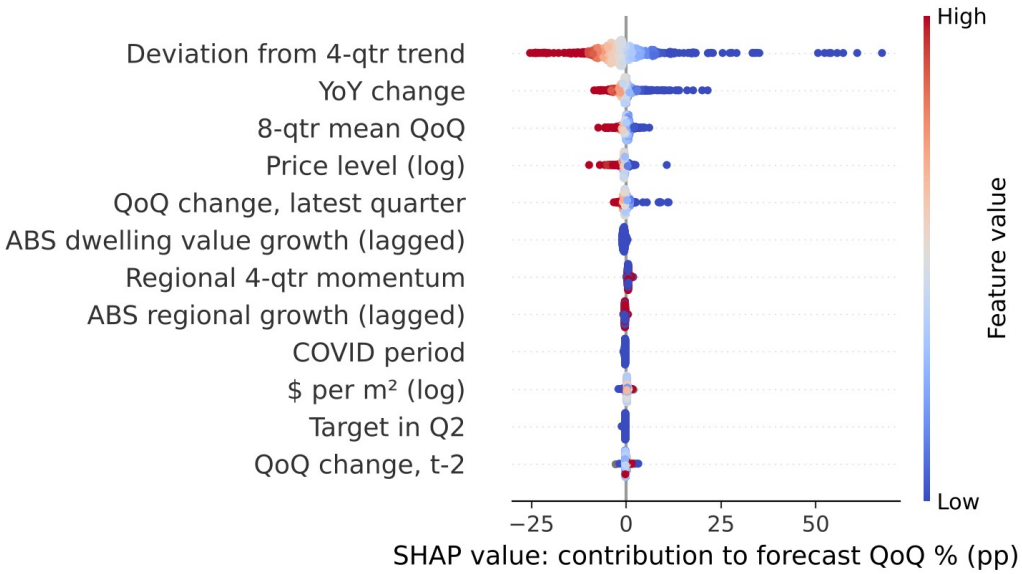


Figure A4.9 — *SHAP summary for LightGBM (h = 1, forecasts for 2026-Q2). Each dot is one suburb; colour is the feature value; position is its push on the forecast.*

3.7 Discussion of Results

Taken together, the evaluation supports four conclusions, and we think each is more useful than a single accuracy number.

- **Pooling across suburbs works, and it is the main source of skill.** The pooled models beat per-suburb SARIMA and Prophet decisively and significantly. That vindicates the A2 decision to move from per-suburb to pooled modelling.
- **Beyond one quarter, suburbs follow their region.** Suburb-specific predictability is short-lived and largely mean-reversion. For two- to four-quarter forecasts, the defensible output is the regional trend plus a wide, honest band, which is exactly what the app now shows.
- **The ceiling is measurement, not modelling.** Five well-tuned model families land within 0.1 pp of each other, and error tracks sampling noise almost perfectly. The next real gain will come from better price measurement, not a bigger model.
- **Calibrated uncertainty is achievable and cheap.** CQR delivers close to nominal coverage at every horizon and in every segment for the cost of two extra LightGBM fits.

Against the A1 problem statement, *can public sales data plus ABS data predict dwelling price changes one to four quarters ahead with acceptable accuracy?*, our answer is therefore qualified. **Yes for one quarter, and for direction at the regional level; only as a calibrated range for two to four quarters at suburb level.** We would rather state that clearly than present a four-quarter point forecast with false confidence.

3.8 Deployment

A2 ended with a working app that ran only on a team member's laptop. For A4 we treated deployment as an engineering problem with three separate concerns: what the app is allowed to see, how it is packaged, and how it stays current.

3.8.1 The Web Application (v4)

The app was rebuilt around the forecasting output rather than the back-test tables. Its main page (Figure A4.10) shows, for any of the 1,279 suburbs with enough recent sales, the observed median price history, a

four-quarter central forecast, the calibrated P10–P90 band as a shaded fan, and a reliability label (high/medium/low) based on recent sales volume and band width. Below the chart, a plain-language note states the typical back-tested error and explains *why* a suburb median can jump for reasons unrelated to any one home's value. We added it because, during our own review of the A2 app, we kept reading large forecast swings as market calls when they were usually a change in the mix of homes sold. Three further tabs cover the regional market, model accuracy and data lineage.

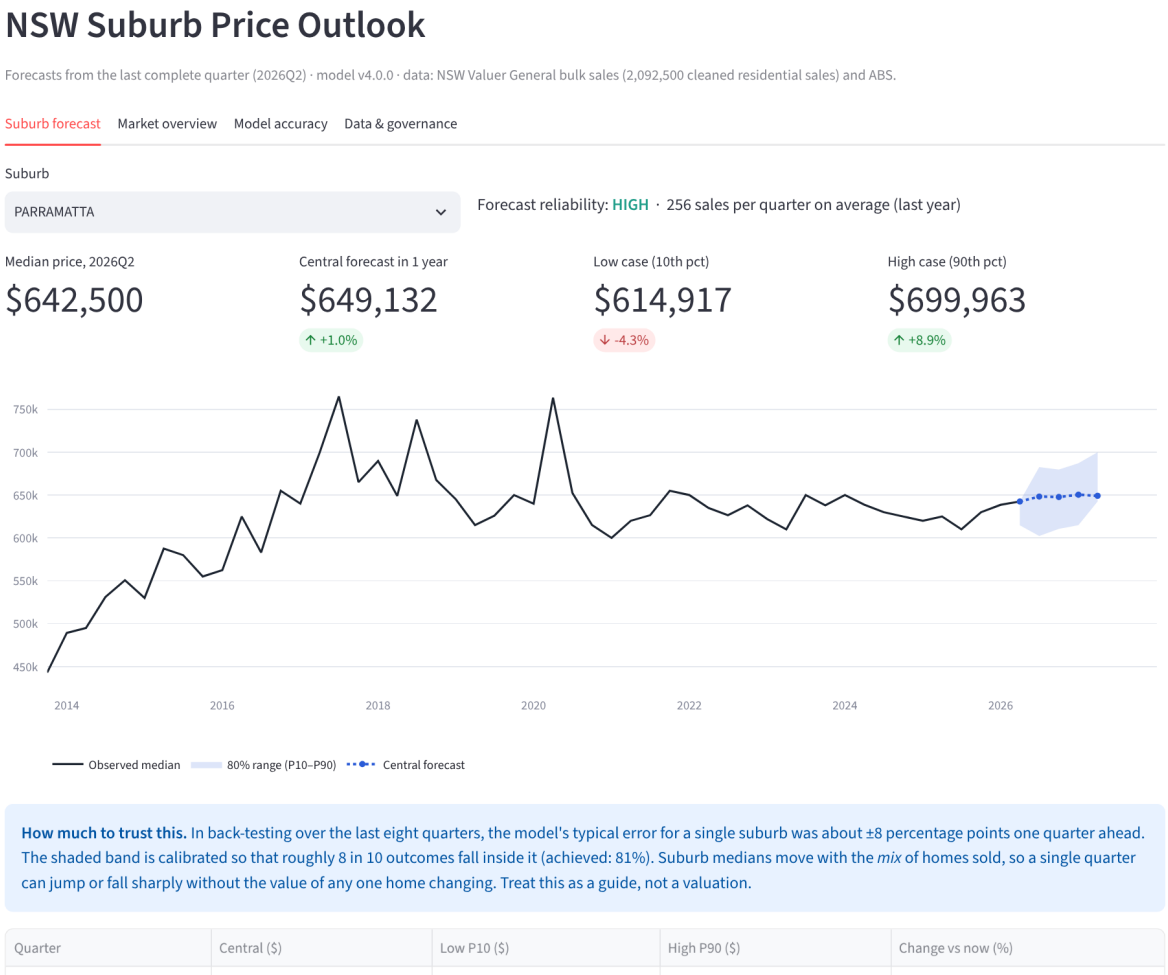


Figure A4.10 — *Web application v4, Suburb Forecast page (live screenshot, Parramatta). The shaded fan is the conformalised P10–P90 band from §3.5.4.*

A deliberate design decision is that **the app never connects to the database or to sale-level data**. A publishing step (`scripts/build_app_data.py`) exports five small, already-aggregated files to `app/data/` (0.57 MB in total), applying the five-sale suppression rule on the way out. This shrinks the attack surface to near zero, makes the app start in about two seconds, and means a deployed copy cannot leak an address even if misconfigured.



Figure A4.11 — *Model Accuracy tab: the same leaderboard and horizon comparison reported in §3.6, generated from the back-test tables.*

3.8.2 Packaging, Continuous Integration and Scheduling

Table A4.13 — *Deployment artefacts.*

Artefact	What it does	Status
`Dockerfile`	Python 3.11-slim image with only the five app dependencies; runs as a non-root user; health check on Streamlit's `/_stcore/health`	Written and reviewed
`requirements-app.txt`	Minimal runtime for Streamlit Community Cloud (the A2-recommended host)	Tested locally
`.github/workflows/ci.yml`	On every push/PR: install pinned deps, lint with ruff, run the test suite, apply both schema files to a throw-away PostgreSQL 16 service (twice, to prove idempotence)	Ready to activate on push
`.github/workflows/weekly_refresh.yml`	Every Tuesday 06:00 AEDT (after the Valuer General's Monday release): acquire → parse → clean → features → forecast → publish → load DB → test; commits refreshed app data, which triggers a Streamlit Cloud redeploy	Ready to activate on push
`Makefile` / pipeline CLI	`make all` reproduces every number in this report from raw archives	Run end-to-end for this report

The full pipeline run is not free. The statistical back-test dominates (about two hours on two CPU cores, almost all of it SARIMA and Prophet fitting). So the weekly job deliberately **excludes** the back-test: it re-trains and re-forecasts, which takes a few minutes, and the back-test is re-run at each tagged release. This keeps the published model and its quoted accuracy in step without paying for a full evaluation every week.

3.9 Deployment Readiness

We assessed readiness against a short checklist we would apply to any analytics product going in front of non-technical users (Table A4.14). Our honest summary: the system is **ready for a supervised public pilot** but not for unsupervised production use. The two gaps are organisational rather than technical: the

workflows have to be pushed to the team's GitHub to go live, and there is no named owner for the weekly job after the semester ends.

Table A4.14 — *Deployment readiness assessment.*

Criterion	Evidence	Ready?
Reproducible from raw data	Pipeline reproduces A2 Table 3.1 to within one record (§3.4.1)	Yes
Automated tests pass	13 tests, including look-ahead and suppression tests (Figure A4.12)	Yes
Accuracy known and published	Common-window back-test, shown in the app	Yes
Uncertainty calibrated	CQR bands hit 80.4–81.3% coverage against an 80% target	Yes
Privacy controls enforced in code	Suppression view, app reads aggregates only, role tests (Figure A4.3)	Yes
Model versioned and traceable	`model_registry` with SHA-256 of each artefact; CHANGELOG	Yes
Monitoring & alerting	Pipeline audit log exists; no automated drift alert yet	Partial
Live public URL	Config complete; requires push to team GitHub + Streamlit Cloud connect (~2 min)	Pending team action
Operational owner after semester	Not assigned	No

```
$ pytest -v
test_parser_reads_nested_archives PASSED
test_cleaning_rules_and_reconciliation PASSED
test_outliers_flagged_not_deleted PASSED
test_grid_is_complete_and_qoq_never_spans_a_gap PASSED
test_partial_quarter_has_no_target PASSED
test_features_do_not_look_ahead PASSED
test_abs_regressor_is_lagged_one_quarter PASSED
test_metrics_known_values PASSED
test_interval_scores_coverage PASSED
test_diebold_mariano_detects_better_model PASSED
test_sample_weight_is_capped_sqrt PASSED
test_full_run_reproduces_a2_table_3_1 PASSED
test_published_forecasts_are_ordered_and_suppressed PASSED
============================== 13 passed in 1.79s ==============================
```

Figure A4.12 — *Automated test suite (actual pytest output). The look-ahead test perturbs all post-origin sales and asserts the features at the origin are unchanged.*

3.10 Governance

Governance in A2 was mostly process: Jira epics, pull requests, a CHANGELOG. In A4 we pushed it into the artefacts themselves, so that the system enforces the rules rather than relying on people to remember them.

- **Model registry.** Every trained model is a row in `model_registry` with its version, horizon, the date it was trained through, its hyper-parameters, its back-test metrics and the SHA-256 hash of the saved file. Whatever produced a published forecast can be identified exactly.
- **Selection discipline.** Hyper-parameters were chosen on a validation window that closes before the test window opens (§3.5.2), and the test window was evaluated once. The champion (LightGBM) was fixed from the validation results, and the rule is recorded in `docs/MODEL_CARD.md`, so the test leaderboard could not quietly steer model choice.
- **Change control.** `docs/CHANGELOG.md` records every data-rule, schema, feature and model change with its effect on reported numbers, including all six audit findings. We recommend branch protection on `main` (one required review, CI must pass) as a repository setting; the steps are in `docs/DEPLOYMENT.md`.
- **Documentation as deliverable.** A model card (intended use, out-of-scope uses, metrics by segment, caveats; after Mitchell et al., 2019), a data-sources log with access dates and licences, and a security and privacy control matrix all ship in `docs/`.

Team process. Sprints 9 and 10 kept the A2 cadence: two-week sprints mirrored as Jira epics, a weekly stand-up on Teams, and a pull request per task with one reviewer. Table A4.15 shows final sprint status.

The ten-sprint plan from A1 is complete except the optional second-state extension, which we deliberately descoped in Sprint 9 to protect the audit and evaluation work.

Table A4.15 — *Final sprint status.*

Sprint	Focus	Status
1–8	Business understanding → data → EDA → models → back-test (A1/A2)	Complete
9	Audit of A2 artefacts; reproducible package; DB v2 and roles; feature store v2; tuning	Complete
10	Common-window back-test; uncertainty calibration; app v4; CI/CD; final report	Complete
Stretch	Second state (VIC) for generalisability	Descoped (see §3.13)

3.11 Ethics, Privacy and Security

The data is public under CC BY 4.0, but "public" is not the same as "harmless". Valuer General records include street addresses and exact prices. Combined with a suburb and a quarter, a thin market can make an "aggregate" effectively disclose one household's sale. Our controls are layered, so that no single mistake exposes anything:

- **Minimum cell size.** No statistic built from fewer than five sales is ever published. This is enforced in the database view, in the app-data export and by a test. It suppresses 55,883 suburb-quarters.
- **Least privilege.** Only the ETL role can read addresses; the app role can read forecasts and the privacy view only (Figure A4.3). The app image contains no credentials and no sale-level data.
- **No false precision.** Every forecast carries a calibrated band and a reliability label, low-liquidity suburbs are labelled *low* rather than hidden, and the app states plainly that it is not a valuation or financial advice.
- **Fairness across places.** We checked whether the model serves regional NSW worse than Sydney. Overall it does not: one-quarter-ahead MAE is 10.6 pp for regional suburbs and 10.9 pp for Sydney, and band coverage is 79.3% and 81.5%. Among liquid suburbs, regional forecasts are actually *more* accurate (6.6 vs 9.5 pp), plausibly because Sydney medians mix houses and apartments far more. Both regions show a small negative bias (forecasts 1.5–1.7 pp below outcomes on average) because the test window was a period of strong growth. We publish these segment results in the model card rather than averaging them away.
- **Secrets and supply chain.** Credentials come from environment variables and GitHub Secrets, `.env` is git-ignored, and all dependencies are pinned to the versions used to produce this report.

One ethical risk we cannot fully engineer away is **misuse as a buy/sell signal**. A forecast that suburb X will rise could, at scale, become self-fulfilling or be used in marketing. We mitigate this with framing (ranges first, central value second; "guide, not valuation"), and the model card lists investment advice and automated lending decisions as out of scope.

3.12 Limitations

- **The noise floor is real.** A suburb median from 20 sales has substantial sampling error, so a large part of each QoQ move is unforecastable composition noise (§3.7). Our forecasts are best read as *direction and range*, not a point estimate.
- **One state.** NSW only. The pipeline is config-driven, but other states publish in different formats and under different terms, so generalisability is untested.
- **Region proxy.** Sydney vs rest-of-NSW is still assigned by postcode range, not the ABS GCCSA correspondence, and suburbs are matched by name, so boundary changes over 2014–2026 are not tracked.
- **No macro drivers.** Interest rates, credit growth and population flows are absent: sourcing, aligning and lagging these series correctly was out of scope for the project window. The model will therefore react to a rate shock only after it shows up in sales.
- **Test window.** Eight quarters (2024-Q3 to 2026-Q2) is a reasonable but single regime. The model has not been tested through a sharp downturn.

3.13 Future Improvements

- **Repeat-sales or hedonic index per suburb.** Tracking the same properties over time (property_id is in the data) would strip out composition noise and could cut the noise floor substantially. This is the single most promising next step.
- **Hierarchical forecasting.** Reconciling suburb, region and state forecasts (e.g. MinT; Hyndman & Athanasopoulos, 2021) so they add up coherently and borrow strength across levels.
- **Macro features and scenario mode.** Add the RBA cash rate and ABS lending indicators, and let users ask "what if rates rise 50 bp?" with the band widening accordingly.
- **Drift monitoring.** Compare each new quarter's realised errors and band coverage with the back-test and alert when they leave the expected range.
- **Second state and SA2 geography.** Victoria's Valuer-General data, and the ABS postcode-to-SA2 correspondence to replace the postcode-range region proxy.

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Section 4 – Individual Work and Contribution Summary

4.1 Ahsan Uddin

Student ID: S386288

Individual Capstone Project Completion Evidence

[INSERT SCREENSHOT: Individual Capstone Project name, completion status, score, and completion date & time]

Individual Project Contribution (since Assessment 2)

I continued as project lead and owned acquisition and automation. I adapted the acquisition script into the package, wrote the weekly scheduled-refresh workflow that runs the pipeline every Tuesday after the Valuer General's release, restructured the repository around `src/`, `tests/` and `docs/`, wrote the CI workflow, and replaced the broken `requirements.txt`. I also ran Sprint 9–10 planning in Jira and the weekly stand-ups.

- Acquisition module · Weekly refresh workflow · CI workflow
- Jira Sprint 9–10 epics · Commit history

Professional Reflection

The most significant work this phase was turning a manual process into a scheduled one. The hard part was not the cron line but making the job safe to run unattended: it had to skip the two-hour back-test, fail loudly, and only publish if the tests passed. Coordinating four people through an audit that found defects in our own earlier work also needed care. We agreed early to treat findings as process issues rather than anyone's fault, and that kept the team moving. My contribution made sure the system keeps working after we stop touching it, which is what moves the project from coursework to something deployable.

4.2 Ferdous Anwar Anik

Student ID: S388272

Individual Capstone Project Completion Evidence

[INSERT SCREENSHOT: Individual Capstone Project name, completion status, score, and completion date & time]

Individual Project Contribution (since Assessment 2)

I led the Sprint 9 audit and the data-engineering rebuild. I wrote the nested-archive parser that reproduces A2's 2,762,903 raw records, corrected the cleaning module so it reproduces A2 Table 3.1, and vectorised outlier flagging (179 s → 5 s). I designed schema v2 (model registry, forecasts, audit log, privacy view), wrote the transactional COPY loader that repopulated `raw_sales` and `suburb_dim`, and implemented the least-privilege roles. I rebuilt the feature store on a complete grid with the lagged ABS regressor, and wrote the look-ahead and reconciliation tests.

- parsing.py · cleaning.py · features.py · db.py
- schema_v2.sql · tests · Jira tasks

Professional Reflection

The most significant thing I did was find that 9.5% of our own QoQ targets spanned a missing quarter. It was an uncomfortable discovery, because the A2 feature store was my work. The fix itself was simple; deciding to rebuild rather than patch, and then proving the new store had no look-ahead, took longer. Writing a test that deliberately corrupts future data taught me more about leakage than any lecture had. Because modelling and the app both sit on this layer, getting it right was what allowed the rest of the team to trust the numbers in this report. It also gave us a reproducible baseline, so any later change can be checked against known counts.

4.3 Mohd Yah Ya Raiyan

Student ID: S390934

Individual Capstone Project Completion Evidence

[INSERT SCREENSHOT: Individual Capstone Project name, completion status, score, and completion date & time]

Individual Project Contribution (since Assessment 2)

I led modelling and evaluation. I set up the validation design (separate tuning and test windows), ran the design and hyper-parameter search, and built the walk-forward back-test that scores 12 models on identical suburb-quarters across four horizons. That included re-running SARIMA and Prophet and reproducing the A2 Gradient Boosting model. I implemented conformalised quantile regression for the P10–P90 bands and the statistical evaluation: Diebold–Mariano tests, the liquidity analysis, segment errors and SHAP.

- backtest.py · models.py · evaluation.py
- Results tables · SHAP script · Jira tasks

Professional Reflection

The most important result I produced is also the most humbling: once models were compared fairly, the gap between them was much smaller than A2 suggested, and most of the remaining error is sampling noise in suburb medians. My main challenge was resisting the urge to keep tuning on the test set. Fixing the validation window first and evaluating the test window once made the final numbers believable, even when they were less impressive than A2's. A second challenge was compute: the SARIMA re-run alone took over two hours, so I had to plan runs carefully. Calibrating the uncertainty bands was the contribution I am proudest of, because it turns an honest-but-noisy forecast into something a user can actually act on.

4.4 Abrar Bin Khaiyum

Student ID: S395453

Individual Capstone Project Completion Evidence

[INSERT SCREENSHOT: Individual Capstone Project name, completion status, score, and completion date & time]

Individual Project Contribution (since Assessment 2)

I rebuilt the web application around forecasts with calibrated bands and reliability labels, and designed the publishing step that exports only aggregated, suppressed data to the app. I wrote the Dockerfile (non-root, health-checked) and the Streamlit Cloud configuration, and produced the model card, deployment guide and README. I consolidated this final report.

- app.py · build_app_data.py · Dockerfile
- Model card · Deployment guide · Jira tasks

Professional Reflection

The most significant work was redesigning the app so that uncertainty comes first: the shaded band, the reliability label and a plain-English accuracy note now sit beside every number. The technical challenge was keeping the app small and safe. Moving it from database access to a set of published files removed both the credential problem and the risk of leaking sale-level data. Writing the report taught me how much easier communication becomes when the evidence is reproducible: every figure here can be regenerated, so I could write with confidence. Working on deployment also showed me that most of the effort in a data product goes into the parts users never see: packaging, privacy rules and clear explanations.